Software Requirement Specification

for

Arena 3

Sports Centre Management System

Version 2.2

Prepared by
Arena3 team

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Record of changes

This record lists every released version of the SRS, newest first. A = added, M = modified, D = deleted.

Date	Version	A/M/D	In charge	Description
2026-10-12	2.2	M	Arena3 team	Brought the figures in line with Learning UML 2.0: merge nodes in the activity diagrams, goal-phrase ovals without UC codes in the use case diagrams, and unnumbered messages in the sequence diagrams. Redrew the context diagram as a level-0 data flow diagram.
2026-10-12	2.1	M	Arena3 team	Consolidated 86 use cases into 44, each with a full specification. Restated the business rules as BR-01 to BR-53, grouped by category and typed by rule type. Redrew the use case, activity and sequence figures they affect.
2026-10-11	2.0	M	Arena3 team	Redrew all 28 UML figures to the FPT template and Learning UML 2.0 rules. Aligned every section with the deployed app.
2026-10-04	1.4.1	A, D	Arena3 team	Added QR gate check-in, promo codes, at-risk list, deposits, fixed weekly bookings, day passes and loyalty points. Removed member debt and court conversion.
2026-09-27	1.4	M	Arena3 team	Traced each requirement to its endpoint or business rule. Added online payment by payOS.
2026-09-20	1.1	A	Arena3 team	Added receipts, customer care requests and the training module (feature flag F4).

I. Overall requirements

1. Product overview

Arena3 is one web application that runs a single sports centre in Ho Chi Minh City, from court booking to the end-of-day till. It serves badminton, basketball and volleyball courts, membership plans, coached classes and the front desk.

Portal	Route	Who uses it	Main purpose
Public site	`/`, `/login`, `/register`	Guest	Browse prices and classes, sign up, sign in
Member portal	`/app`	Member	Book courts, pay, enrol in classes, show the gate pass
Front desk	`/desk`	Receptionist	Serve walk-ins, take payments, run the gate and the till
Coach portal	`/coach`	Coach	Teach classes, take attendance, track student progress
Manager portal	`/manager`	Manager	Set prices and rules, staff the centre, read reports

The app is built with TanStack Start and React, with a JSON API under `/v1` in the same deployment. Production runs on Vercel with Neon PostgreSQL. Development runs offline on an embedded PGLite database that migrates and seeds itself.

Out of scope. Multi-branch operation, payroll, native mobile apps and SMS delivery are not part of this release. Class fees are covered by membership plans and are not charged per session.

2. Context diagram

The context diagram shows Arena3 as one process exchanging data with five human actors and three external systems.

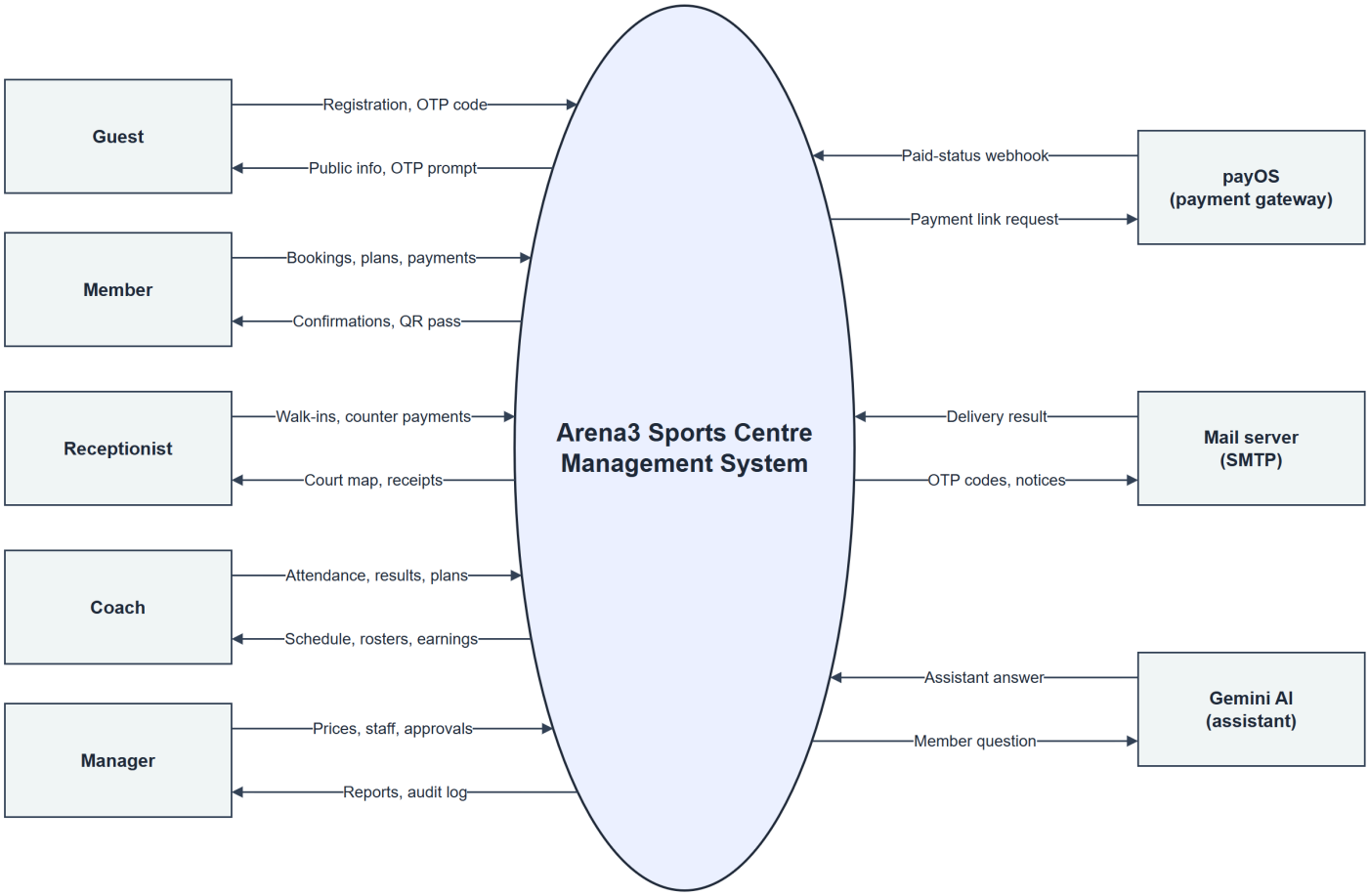


Figure 1. *Context diagram of the Arena3 Sports Centre Management System*

External entity	Data into Arena3	Data out of Arena3
Guest	Registration details, OTP code	Public schedule, prices, OTP prompt
Member	Booking, enrolment and plan requests, online payments	Booking confirmations, QR pass, notifications
Receptionist	Walk-ins, check-ins, counter payments, refund requests	Court map, payment queue, receipts
Coach	Attendance, results, training plans	Teaching schedule, rosters, earnings
Manager	Prices, plans, staff, settings, refund approvals	Reports, audit log, e-invoices
payOS	Signed "paid" webhook	Payment link request
Mail delivery (SMTP)	Delivery result	OTP codes, reminders, notices
Gemini AI	Assistant answer	Member question with centre context only

3. Main workflows

Four activity diagrams cover the workflows that move money or change a schedule. Each uses swimlanes, one initial node, guarded decisions and one final node.

3.1 Register a member account

A guest becomes a member only after proving the email address with a one-time code.

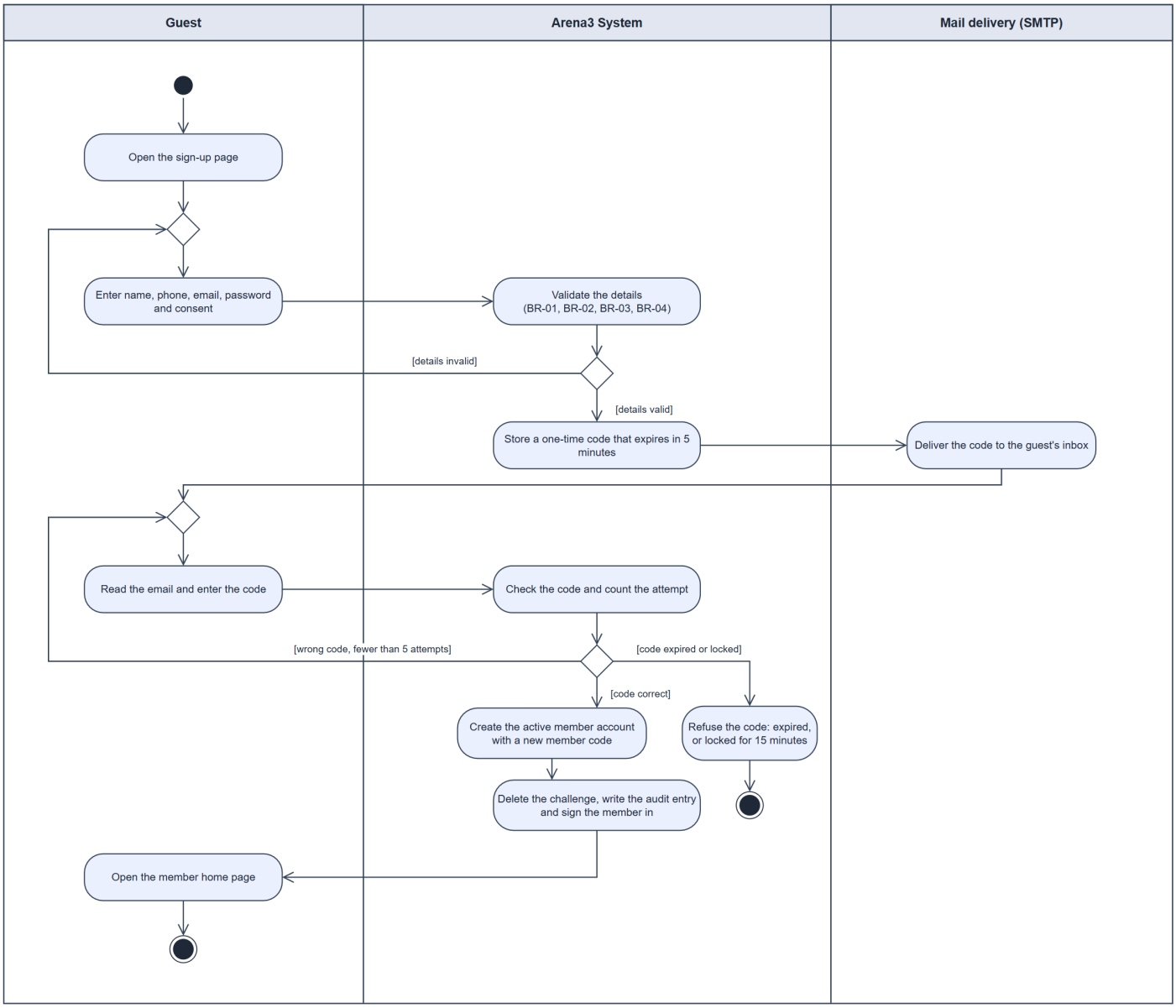


Figure 2. *Activity diagram: Register a member account*

The code expires after 5 minutes. Too many wrong attempts lock the challenge for 15 minutes. A successful check creates the account, writes an audit entry and signs the member in.

3.2 Book a court and pay

A booking is confirmed only when money is known to have arrived or plan hours are spent.

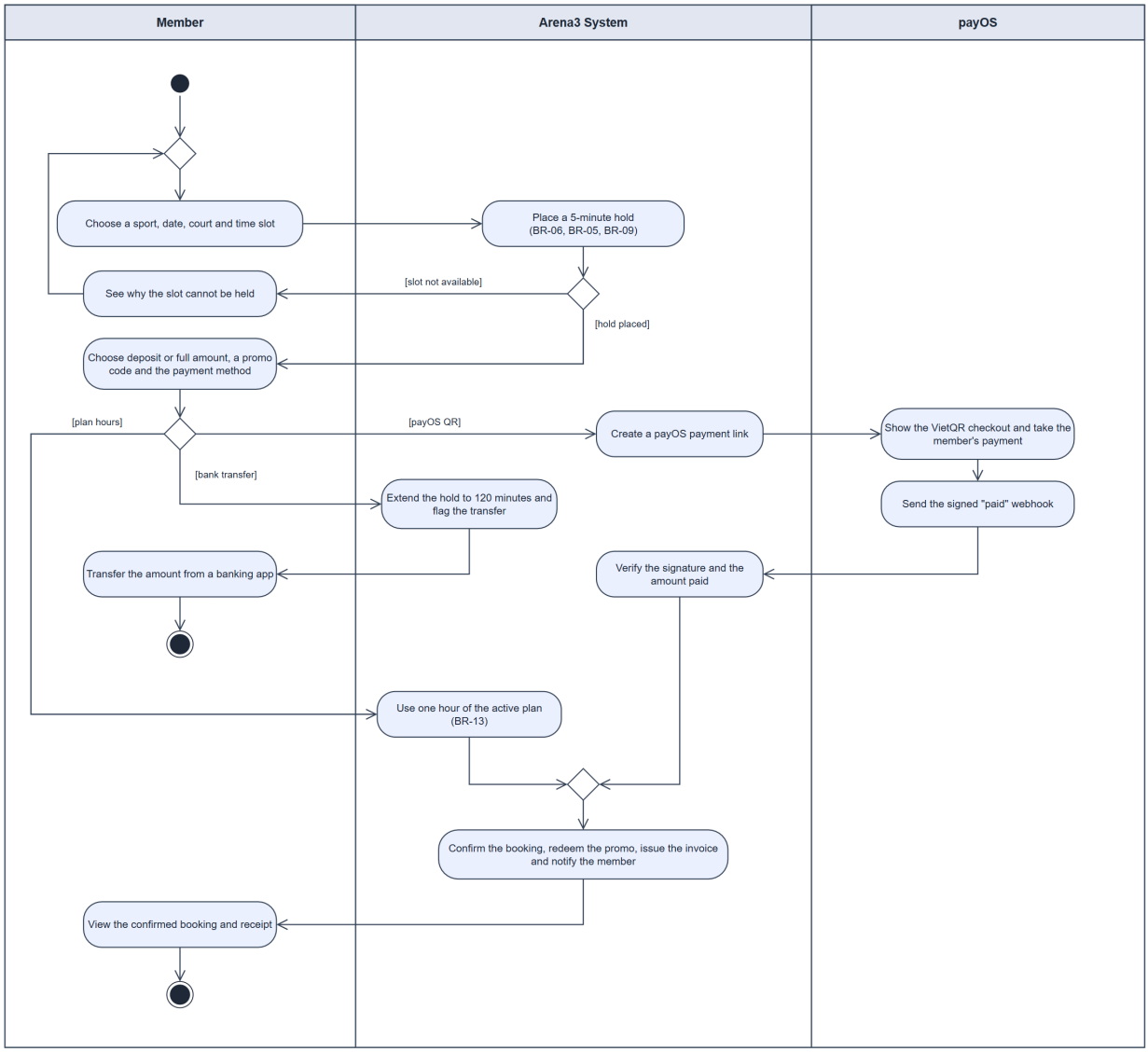


Figure 3. *Activity diagram: Book a court and pay*

A slot is held for 5 minutes while the member pays. Three paths follow:

- **payOS QR:** the booking is confirmed only after Arena3 verifies payOS's signed result and the amount.
- **Bank transfer:** the hold stretches to 120 minutes and the transfer waits for the front desk to reconcile it.
- **Plan quota:** one court hour is taken from the member's active plan.

3.3 Enrol in a class or join the waitlist

A full class never refuses a member outright; it offers a waitlist place instead.

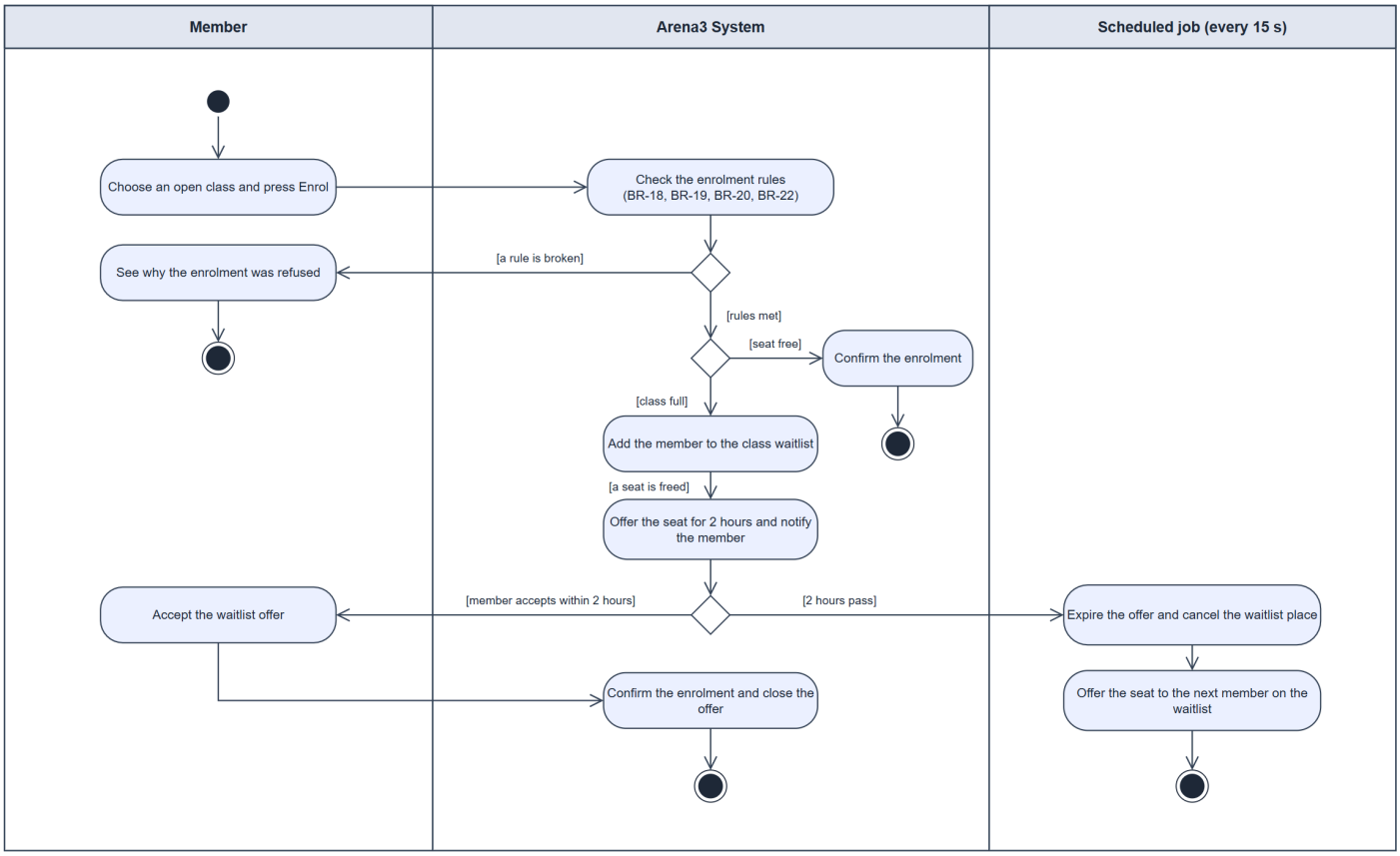


Figure 4. *Activity diagram: Enrol in a class or join the waitlist*

When a seat frees up, the next member gets a 2-hour offer. The scheduled job expires an unanswered offer and moves on to the next member.

3.4 Refund a payment

A large refund needs a second person, so no receptionist can refund their own sale above the limit.

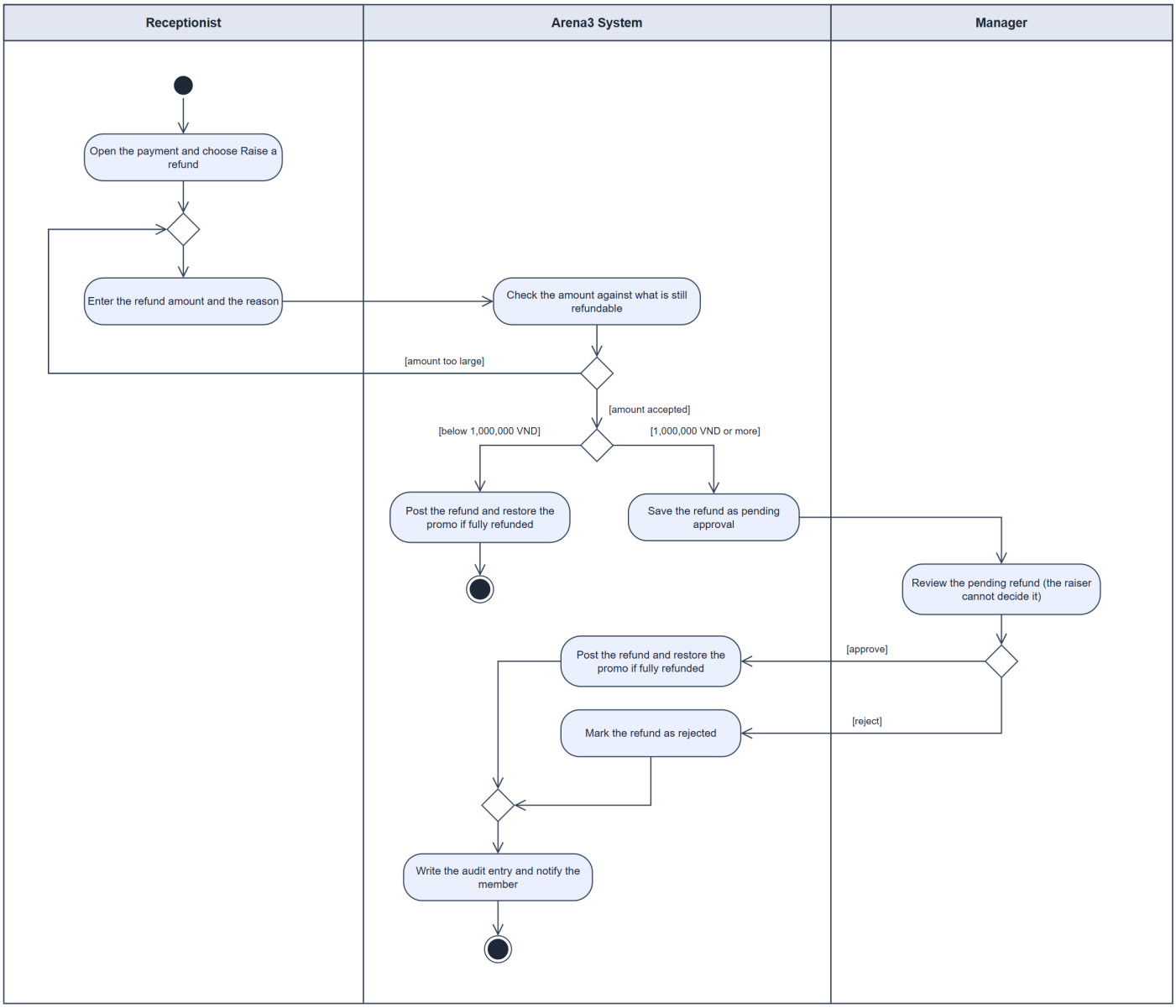


Figure 5. *Activity diagram: Refund a payment*

Refunds up to 1,000,000đ post at once. Larger refunds wait for a manager, who approves or rejects them. Both outcomes are audited and the member is notified.

II. User requirements

1. Actors

Arena3 has five human actors and three external system actors. The four staff and member roles specialise one abstract Registered User, who can sign in.

#	Actor	Kind	Description
1	Guest	Human	Anyone not signed in. Browses prices and classes and registers a member account.
2	Registered User	Abstract	Any signed-in person. Owns the sign-in, profile, password and notification use cases.
3	Member	Human	A customer with an account. Books courts, buys plans, joins classes and enters with a QR pass.
4	Receptionist	Staff	Runs the front desk and the till: walk-ins, payments, gate, gear, refunds up to the limit.
5	Coach	Staff	Teaches classes. Takes attendance and records results, plans, homework and reviews.
6	Manager	Staff	Sets prices, plans, classes and rules. Issues staff accounts, approves large refunds and reads reports.
7	payOS	System	Vietnamese payment gateway. Issues VietQR payment links and sends signed "paid" webhooks.
8	Mail delivery (SMTP)	System	Sends one-time codes, reminders and notices by email.
9	Gemini AI	System	Google's model. Answers member questions and drafts training plans from centre data only.

Staff accounts are issued by a manager; there is no public staff sign-up and no admin role.

2. Use case diagrams

Each diagram covers one feature area inside one system boundary. Every oval is a complete goal of its actor, named by a short verb phrase; the use case list (II.3) gives each goal its UC number, which the specifications (III.2) and the activity and sequence diagrams use. Steps such as verifying a one-time code or paying by payOS QR are flows inside their use case, so the diagrams need no «include» or «extend» relationships. Associations carry no arrowheads.

2.1 Account and access

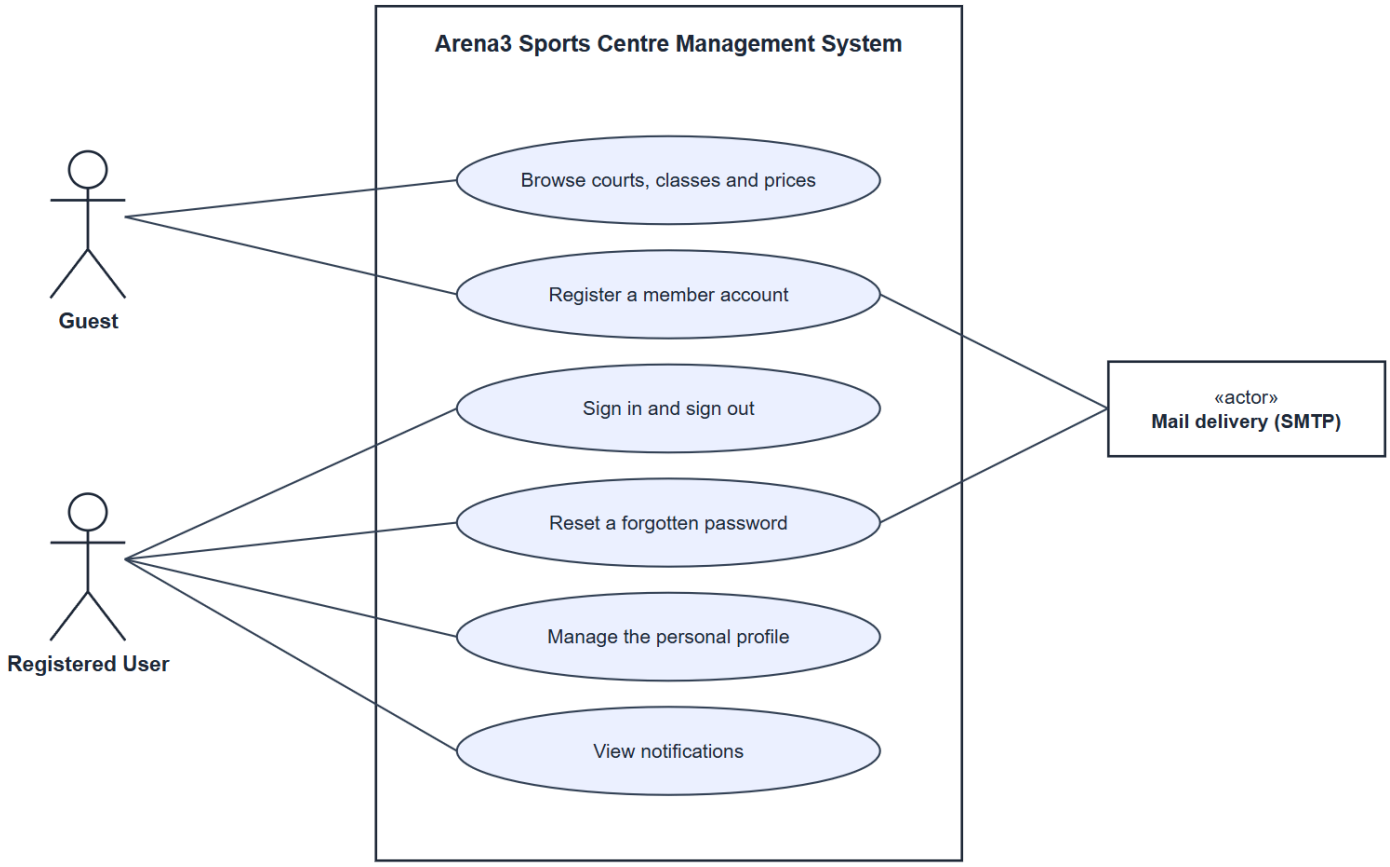


Figure 6. *Use case diagram: Account and access*

2.2 Member: courts, plans and payment

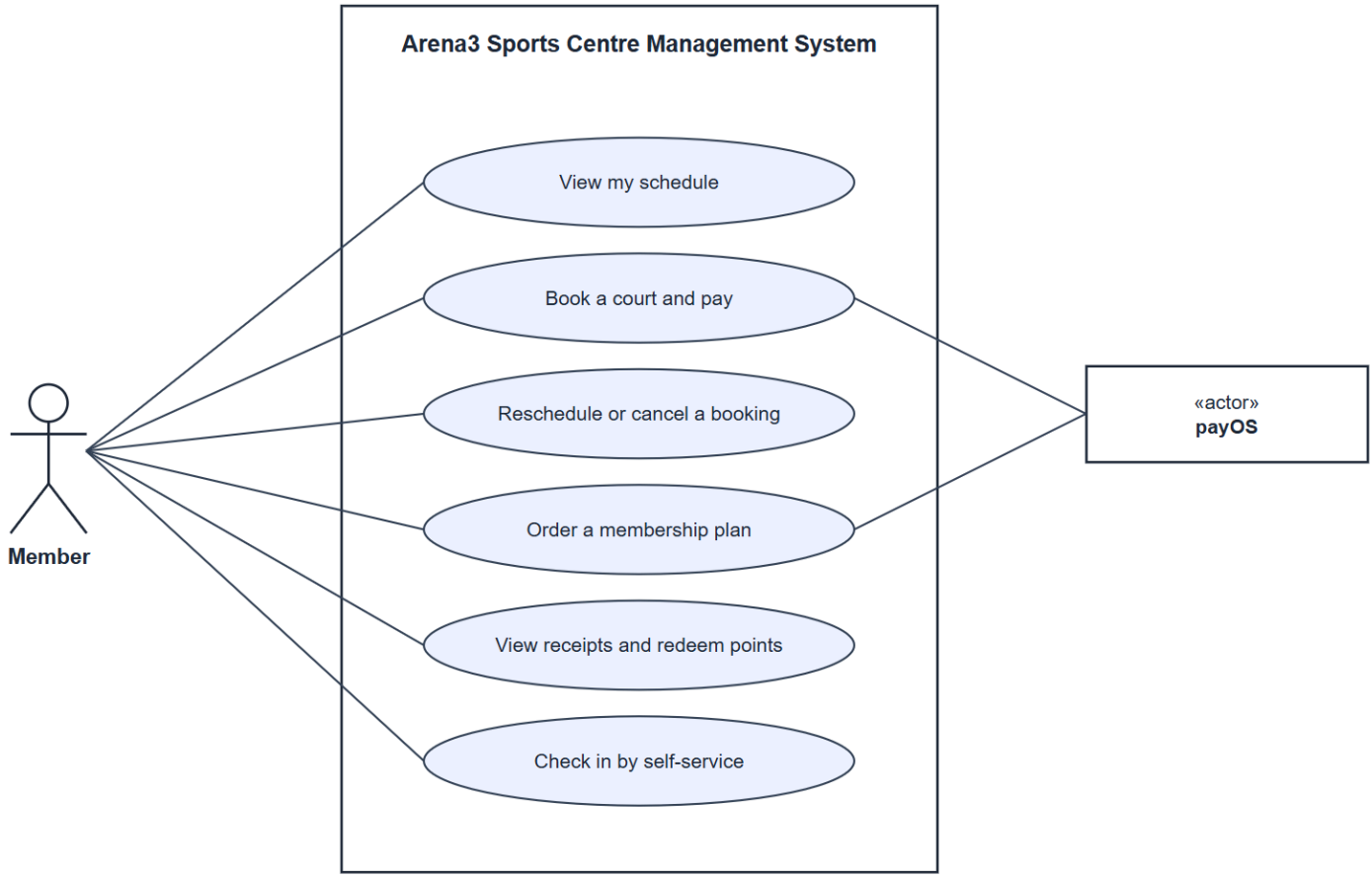


Figure 7. *Use case diagram: Member court booking and payment*

2.3 Member: classes and training

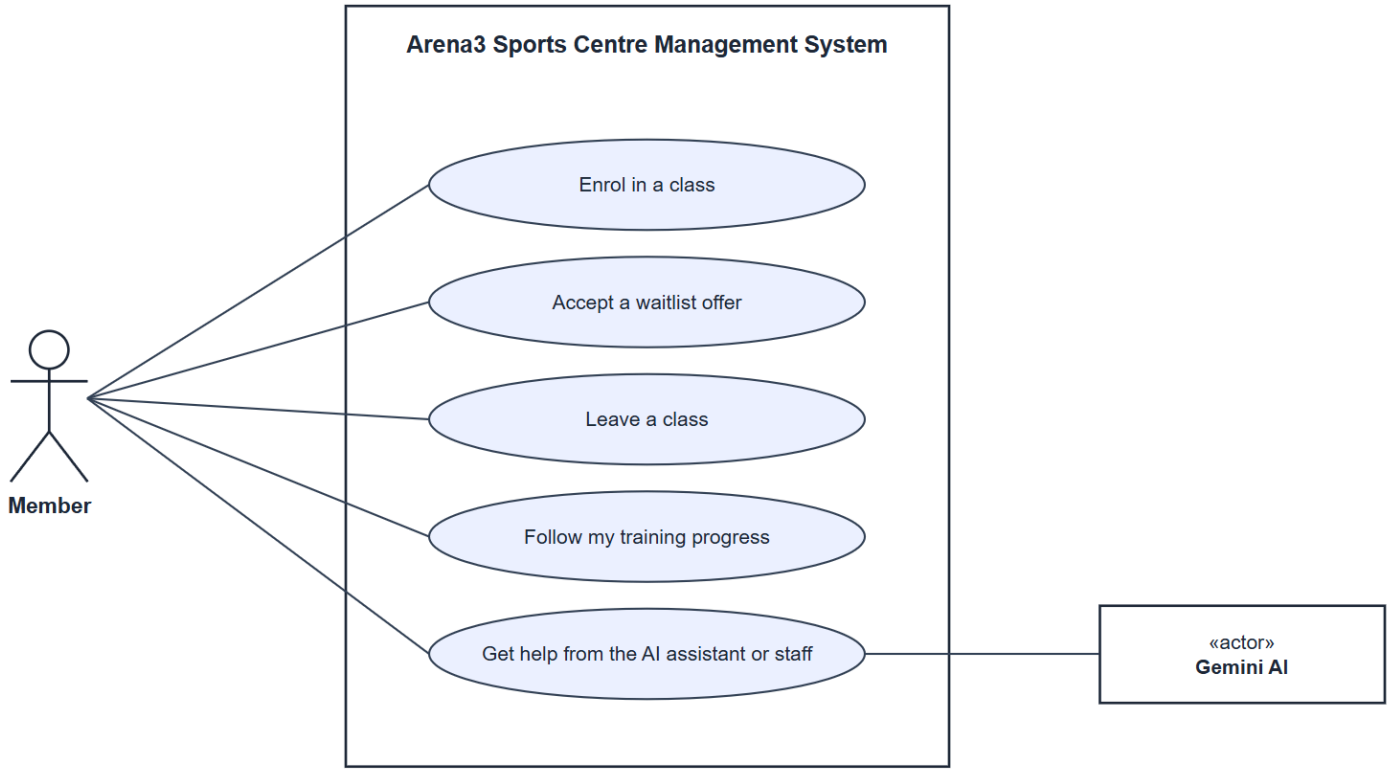


Figure 8. *Use case diagram: Member classes and training*

2.4 Receptionist: sales and the till

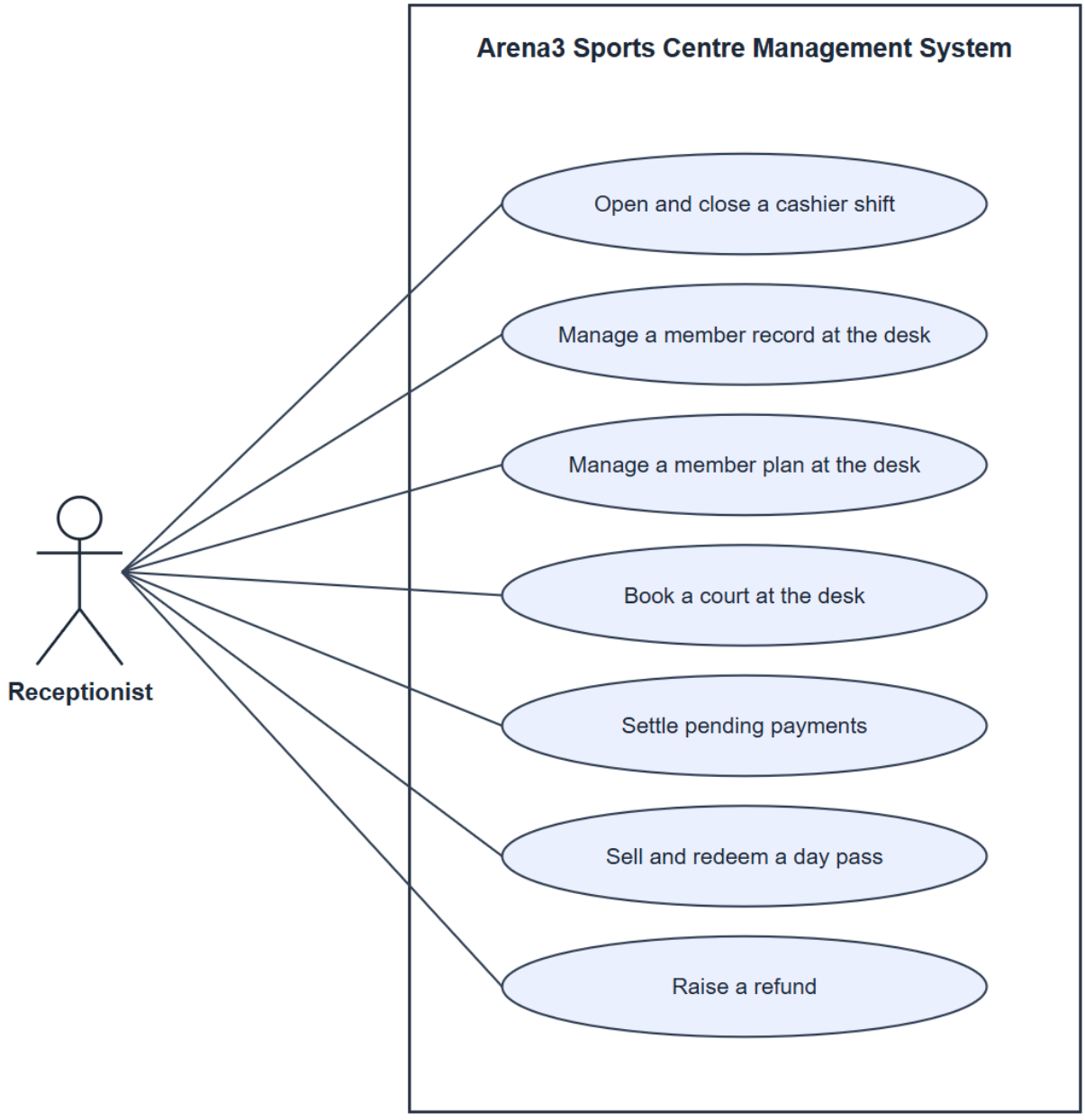


Figure 9. *Use case diagram: Front desk sales*

2.5 Receptionist: daily operations

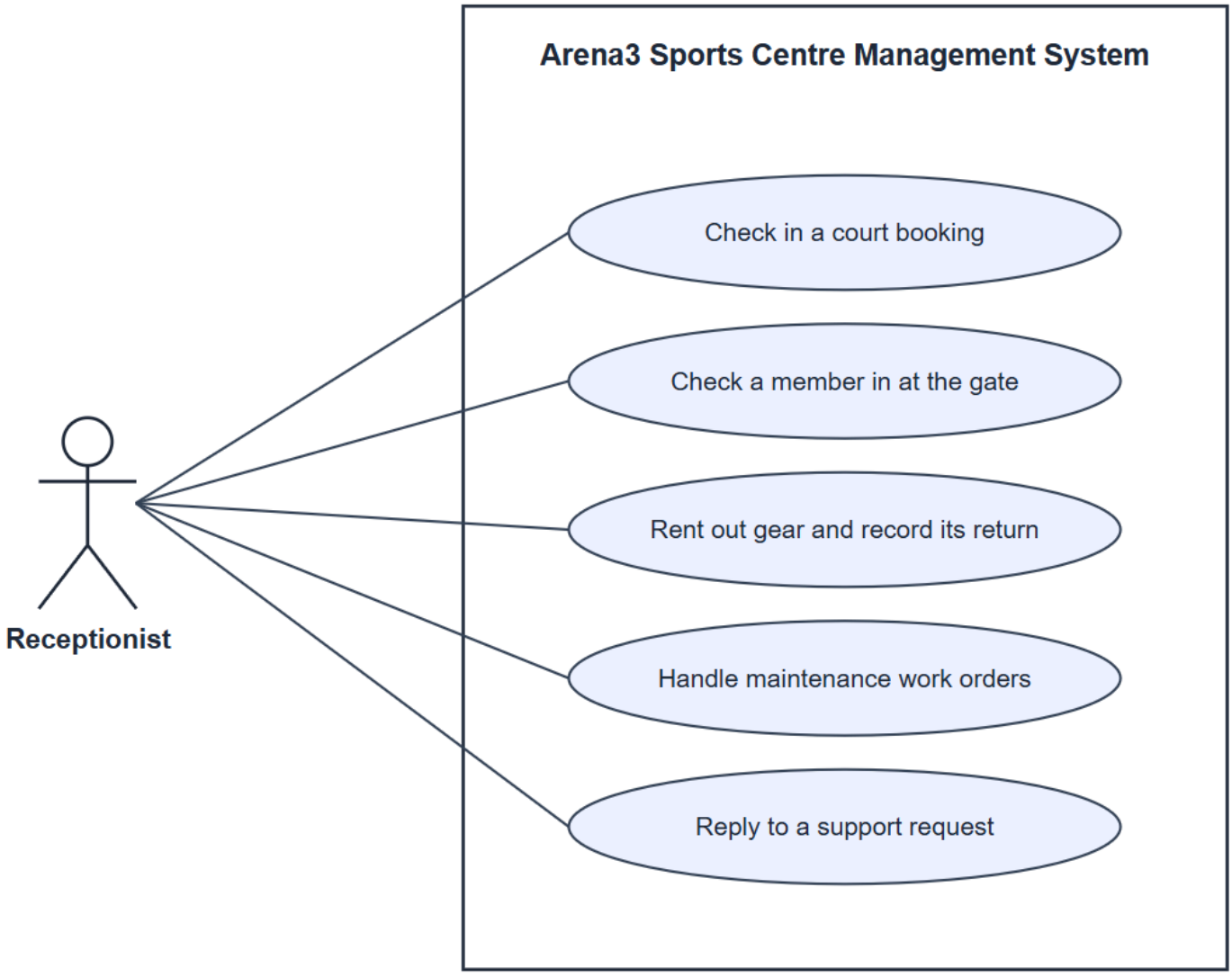


Figure 10. *Use case diagram: Front desk operations*

2.6 Coach

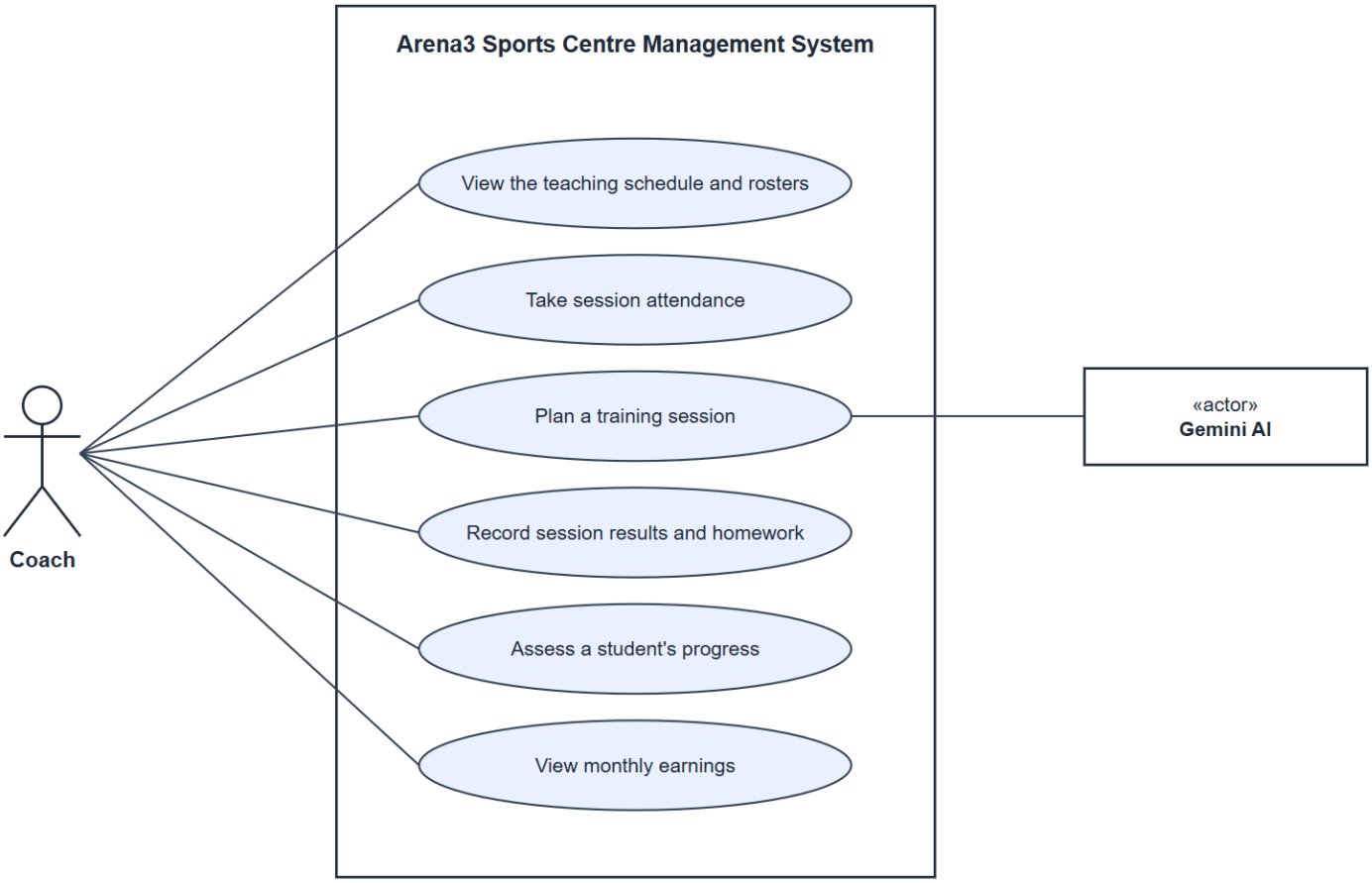


Figure 11. *Use case diagram: Coach*

2.7 Manager: setup

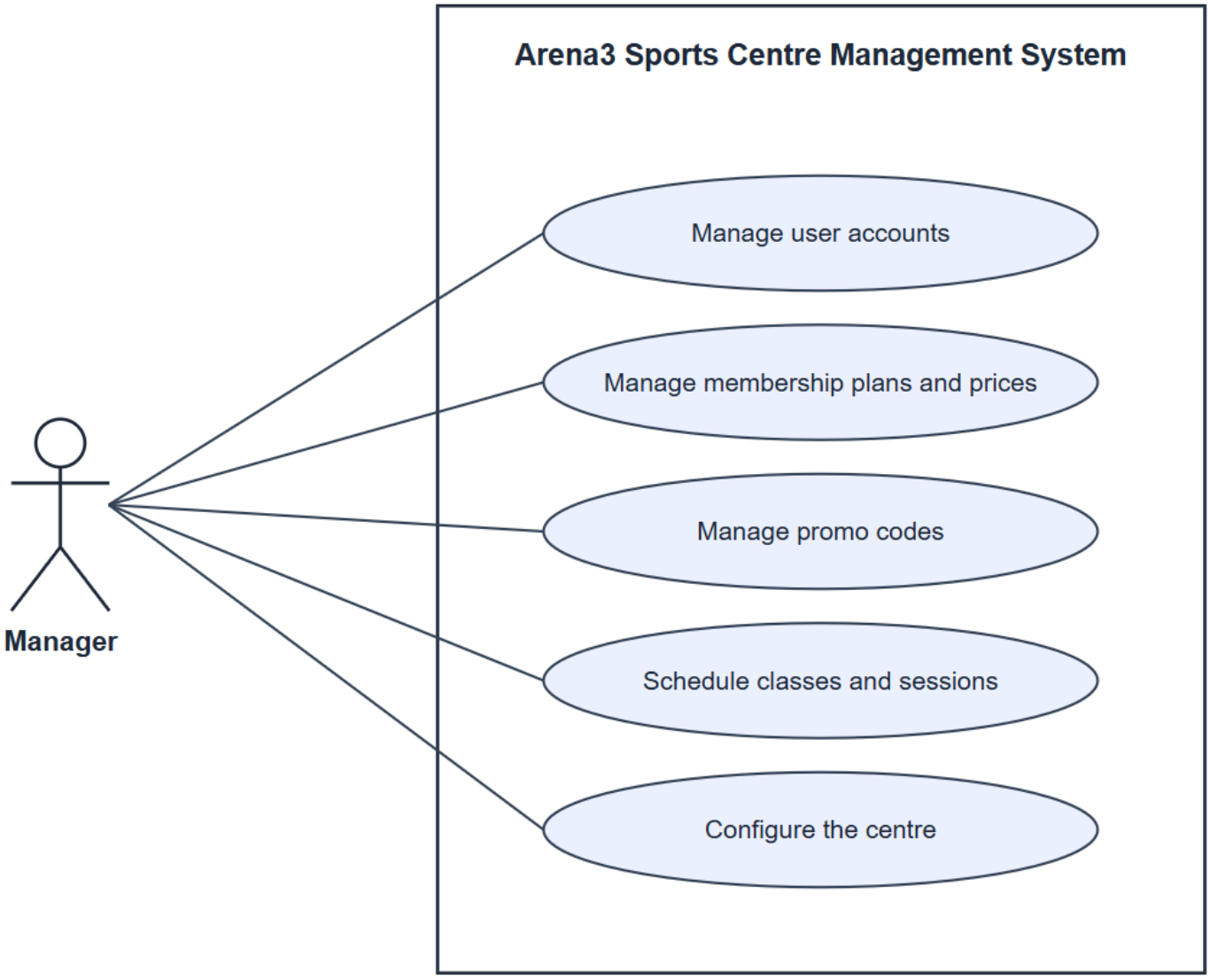


Figure 12. *Use case diagram: Manager setup*

2.8 Manager: oversight

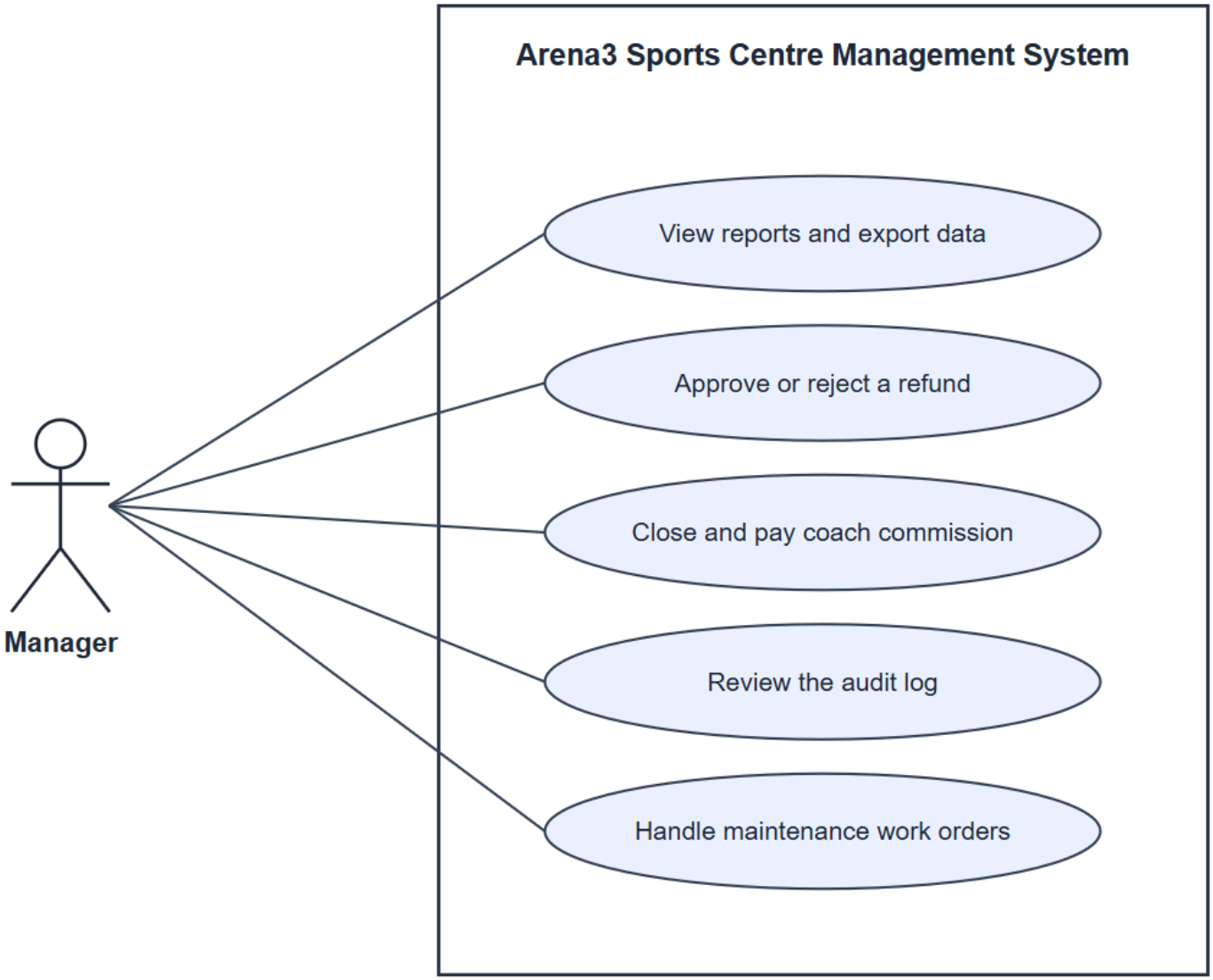


Figure 13. *Use case diagram: Manager oversight*

3. Use case list

The list numbers all 44 use cases in figure order. Each is one complete goal of its primary actor and has a full specification in III.2. Version 2.1 folded the 86 cases of v2.0 into these 44: sub-steps and variants of the same goal became flows of that goal.

ID	Use case	Primary actor	Figure
UC-01	Browse courts, classes and prices	Guest	6
UC-02	Register a member account	Guest	6
UC-03	Sign in and sign out	Registered User	6
UC-04	Reset a forgotten password	Registered User	6
UC-05	Manage the personal profile	Registered User	6
UC-06	View notifications	Registered User	6
UC-07	View my schedule	Member	7
UC-08	Book a court and pay	Member	7
UC-09	Reschedule or cancel a booking	Member	7
UC-10	Order a membership plan	Member	7
UC-11	View receipts and redeem points	Member	7
UC-12	Check in by self-service	Member	7
UC-13	Enrol in a class	Member	8
UC-14	Accept a waitlist offer	Member	8
UC-15	Leave a class	Member	8
UC-16	Follow my training progress	Member	8
UC-17	Get help from the AI assistant or staff	Member	8
UC-18	Open and close a cashier shift	Receptionist	9
UC-19	Manage a member record at the desk	Receptionist	9
UC-20	Manage a member plan at the desk	Receptionist	9
UC-21	Book a court at the desk	Receptionist	9
UC-22	Settle pending payments	Receptionist	9
UC-23	Sell and redeem a day pass	Receptionist	9
UC-24	Raise a refund	Receptionist	9
UC-25	Check in a court booking	Receptionist	10
UC-26	Check a member in at the gate	Receptionist	10
UC-27	Rent out gear and record its return	Receptionist	10
UC-28	Handle maintenance work orders	Receptionist, Manager	10
UC-29	Reply to a support request	Receptionist	10
UC-30	View the teaching schedule and rosters	Coach	11
UC-31	Take session attendance	Coach	11
UC-32	Plan a training session	Coach	11
UC-33	Record session results and homework	Coach	11

ID	Use case	Primary actor	Figure
UC-34	Assess a student's progress	Coach	11
UC-35	View monthly earnings	Coach	11
UC-36	Manage user accounts	Manager	12
UC-37	Manage membership plans and prices	Manager	12
UC-38	Manage promo codes	Manager	12
UC-39	Schedule classes and sessions	Manager	12
UC-40	Configure the centre	Manager	12
UC-41	View reports and export data	Manager	13
UC-42	Approve or reject a refund	Manager	13
UC-43	Close and pay coach commission	Manager	13
UC-44	Review the audit log	Manager	13

III. Functional requirements

1. System functional overview

1.1 Screens flow

Each portal has one screen flow that starts at its entry point and reads left to right. Solid boxes are routed screens; dashed boxes are dialogs or in-page steps; grey boxes are portal homes drawn in their own flow.

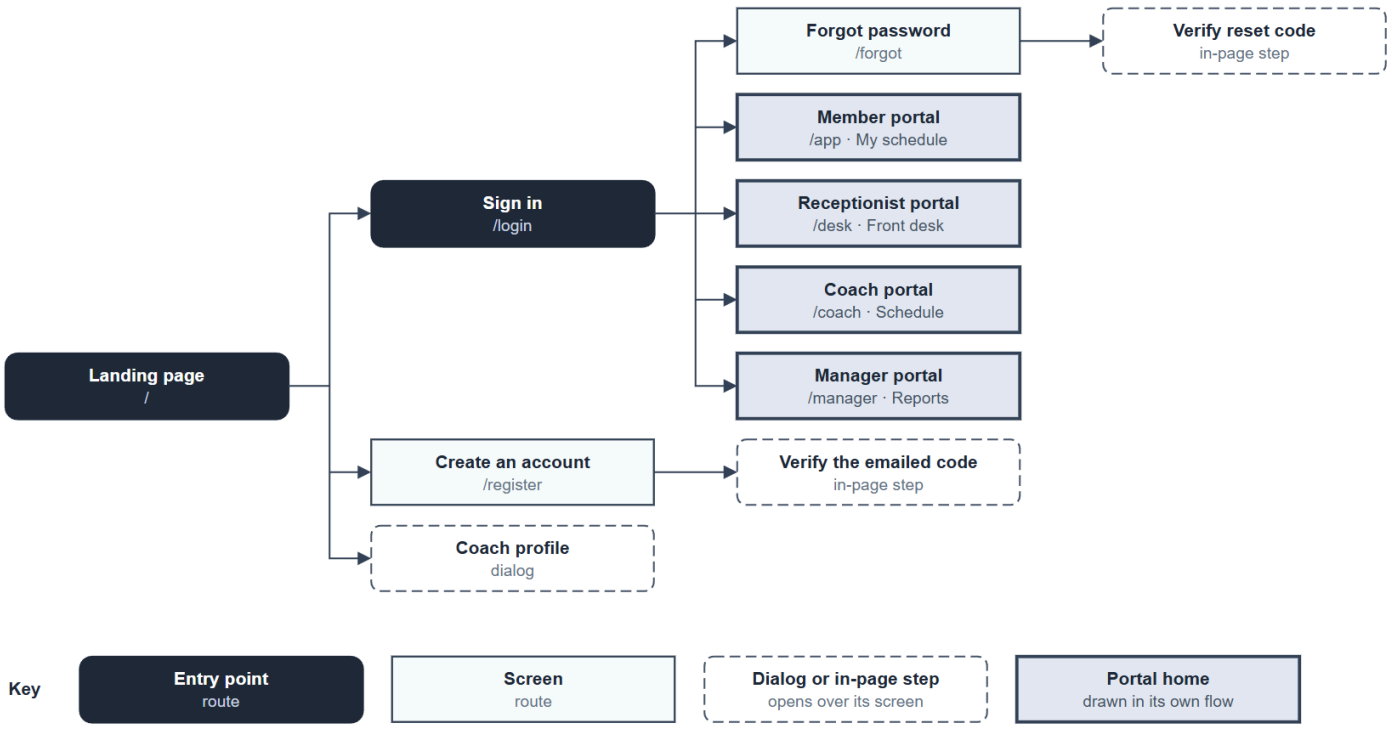


Figure 14. *Screens flow: public site and sign-in*

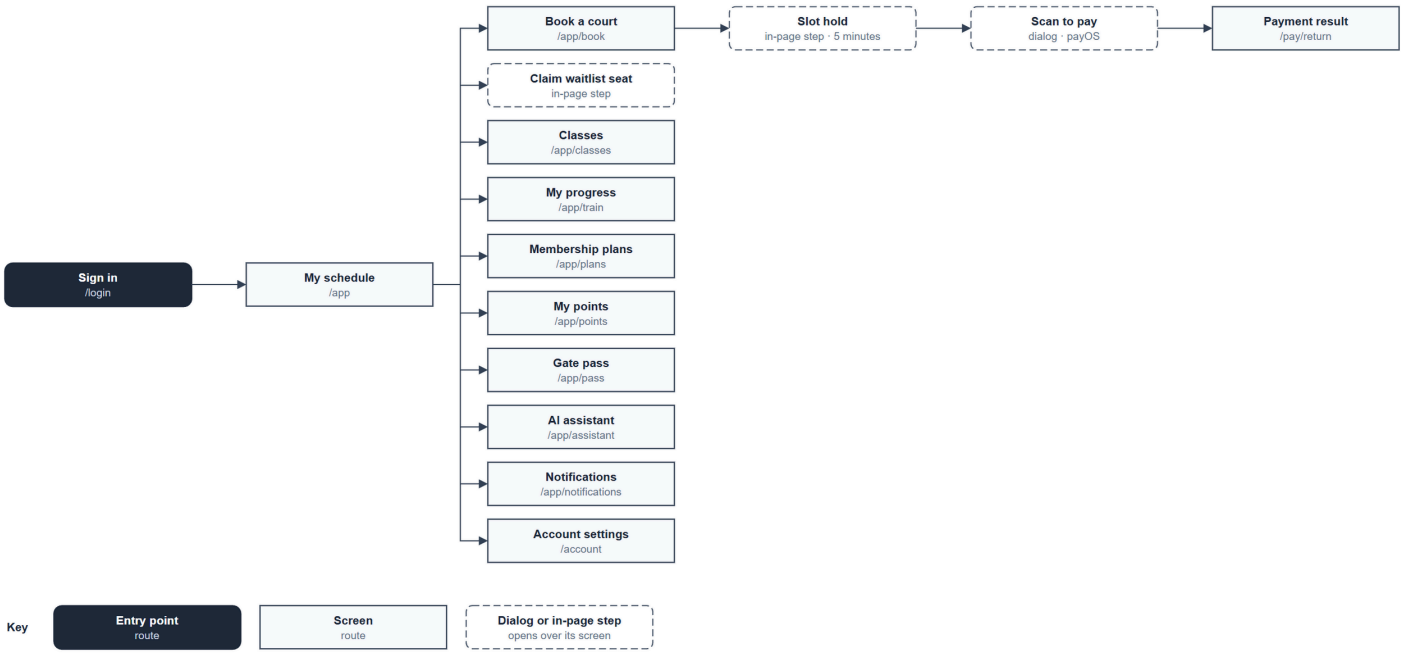


Figure 15. *Screens flow: member portal*

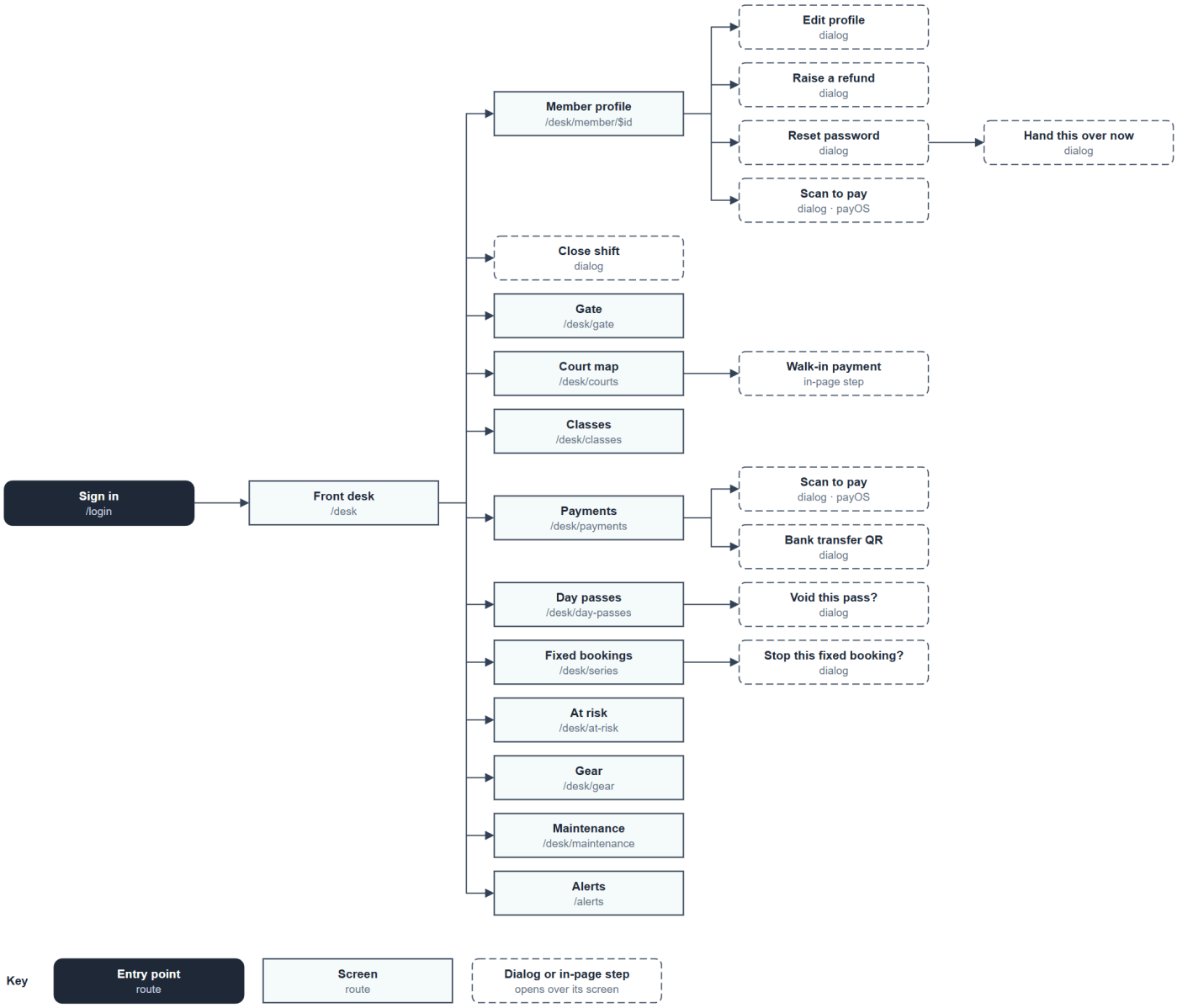


Figure 16. *Screens flow: front desk (receptionist)*

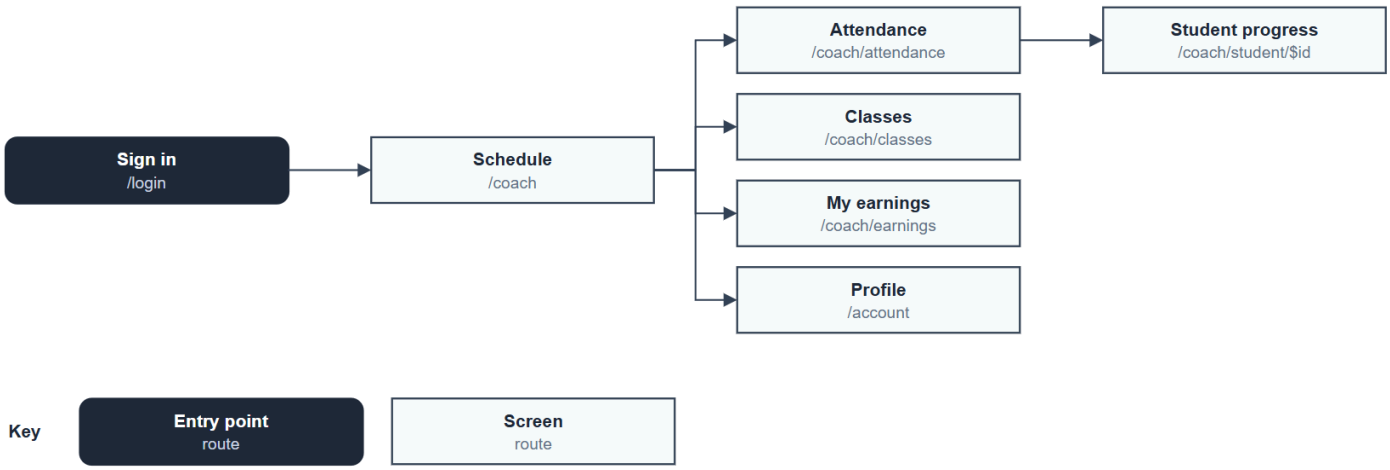


Figure 17. *Screens flow: coach portal*

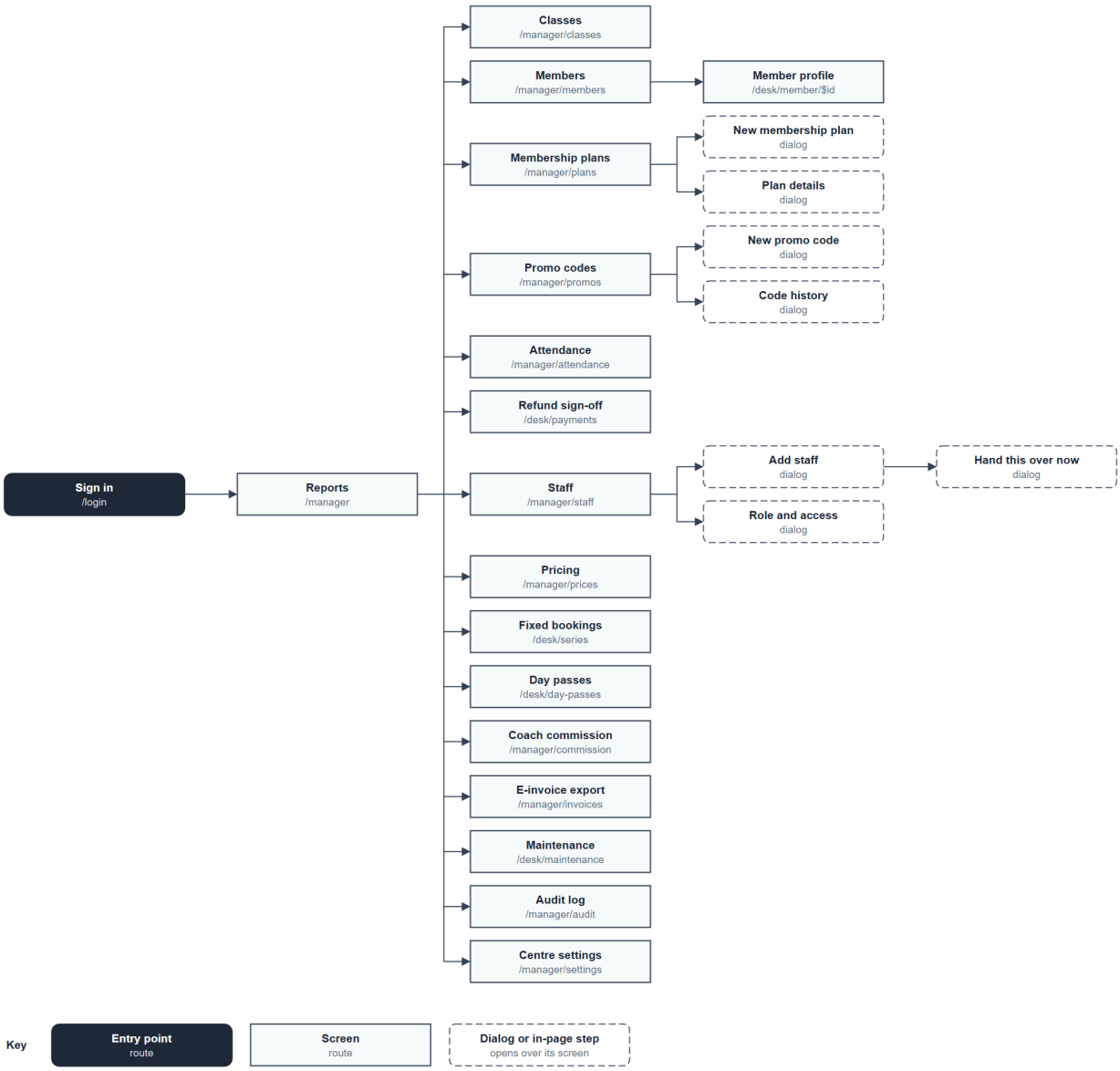


Figure 18. *Screens flow: manager portal*

1.2 Screen descriptions

The app has 44 routed screens. Each row names the route exactly as the app serves it.

#	Feature	Screen	Route	Description
1	Public	Landing page	`/`	Centre introduction, sports, prices, open classes and coach profiles.
2	Account	Sign in	`/login`	Sign in by phone or email and password. Each role lands on its own portal.
3	Account	Create an account	`/register`	Member sign-up with consent, confirmed by a one-time code.
4	Account	Forgot password	`/forgot`	Sends a reset code and sets a new password.
5	Account	Account settings	`/account`	Personal details, password, receipts and support requests.
6	Account	Alerts	`/alerts`	Staff alerts, such as an absence streak.
7	Payment	Payment result	`/pay/return`	Return page after payOS. Asks Arena3 to verify the payment, then shows the outcome.
8	Member	My schedule	`/app`	Upcoming bookings and classes, plan balance and waitlist offers.
9	Booking	Book a court	`/app/book`	Pick a date and sport, tap a free slot, hold it for 5 minutes and pay.
10	Classes	Classes	`/app/classes`	Enrol by sport. A full class offers a first-come waitlist.
11	Training	My progress	`/app/train`	Attendance, results, reviews, training plans and homework.
12	Plans	Membership plans	`/app/plans`	Order a plan here and pay at the desk.
13	Loyalty	My points	`/app/points`	Points balance and history.
14	Gate	Gate pass	`/app/pass`	QR pass whose code changes every minute.
15	Assistant	AI assistant	`/app/assistant`	Gemini answers grounded in the timetable, plans and coaches.
16	Notifications	Notifications	`/app/notifications`	In-app notices for the member.
17	Desk	Front desk	`/desk`	Find a member, sell a plan, take a payment, open and close the shift.
18	Desk	Member profile	`/desk/member/$id`	Plans, bookings and payments. Edit, refund, reset password, take a payOS payment.
19	Gate	Gate	`/desk/gate`	Scan a member's QR pass, or look the member up by phone.
20	Courts	Court map	`/desk/courts`	Live court grid. Sells a walk-in slot.
21	Classes	Classes (desk)	`/desk/classes`	Every class with its sessions, court, coach and roster.
22	Payments	Payments	`/desk/payments`	Plans awaiting payment, transfers to reconcile, refunds awaiting a manager, receipts.
23	Day passes	Day passes	`/desk/day-passes`	Sell, mark as used or void a day pass.
24	Bookings	Fixed bookings	`/desk/series`	Fixed weekly bookings over 4–12 weeks at 10% off.
25	Care	At risk	`/desk/at-risk`	Members whose attendance has dropped, with a contact log.

#	Feature	Screen	Route	Description
26	Gear	Gear	`/desk/gear`	Lend and take back equipment against stock.
27	Maintenance	Maintenance	`/desk/maintenance`	Work orders. An open order can block a court.
28	Coach	Schedule	`/coach`	The coach's own sessions by day, with court, class code and enrolment.
29	Coach	Attendance	`/coach/attendance`	Session register: Present, Late, Absent or Excused, plus results.
30	Coach	Student progress	`/coach/student/$id`	Goal, level, attendance totals, reviews, notes and homework.
31	Coach	Classes (coach)	`/coach/classes`	Classes the coach leads or assists, with plans and homework.
32	Coach	My earnings	`/coach/earnings`	Fee per class plus a fee per student present, by month.
33	Reports	Reports	`/manager`	Revenue by day, method, source and shift; capacity and members reports; export.
34	Classes	Classes (manager)	`/manager/classes`	Create, publish, edit and cancel classes. Clashes are blocked.
35	Members	Members	`/manager/members`	Every member with filters, plan and class count.
36	Plans	Membership plans	`/manager/plans`	Create, edit and withdraw plans. A plan is never deleted.
37	Promotions	Promo codes	`/manager/promos`	Create codes and read their redemption history.
38	Attendance	Attendance report	`/manager/attendance`	Attendance across classes, sessions and members.
39	Staff	Staff	`/manager/staff`	Issue, lock and re-role staff accounts; reset passwords.
40	Pricing	Pricing	`/manager/prices`	Price rules by sport, court, day type and hour band. A new price applies only to new sales.
41	Commission	Coach commission	`/manager/commission`	Close a month's coach commission and mark it paid.
42	Invoices	E-invoice export	`/manager/invoices`	Export invoices as a spreadsheet or XML.
43	Audit	Audit log	`/manager/audit`	Audit entries filtered by actor, action, entity and date.
44	Settings	Centre settings	`/manager/settings`	Opening hours, slot length, windows, limits, VAT, rounding and feature flags.

1.3 Screen authorization

Each portal layout checks the signed-in role before it renders. A wrong role is sent to its own home; no session is sent to Sign in.

Screens	Guest	Member	Receptionist	Coach	Manager
Landing page, Sign in, Create an account, Forgot password, Payment result	X	X	X	X	X
Account settings		X	X	X	X
Alerts			X	X	X
Member portal (`/app/*` , screens 8–16)		X			
Front desk (`/desk/*` , screens 17–27)			X		X

Screens	Guest	Member	Receptionist	Coach	Manager
Coach portal (`/coach/*`, screens 28–32)				X	X
Manager portal (`/manager/*`, screens 33–44)					X

The API checks every request again on the server, so hiding a button is never the only control. These actions are narrower than the screen that shows them:

Action	Allowed roles	Refused with
Approve or reject a refund above the limit	Manager	403 FORBIDDEN
Mark a bank transfer as received or rejected	Receptionist, Manager	403 FORBIDDEN
Raise a payOS link for a plan order	Receptionist, Manager	403 FORBIDDEN
Raise a payOS link for a held court	Member (own hold only), Receptionist, Manager	403 FORBIDDEN
Write a register for a session	Coach (own classes only), Manager	403 FORBIDDEN
Correct a closed register	Manager, with a reason	422 BR-35
Read another member's bookings or receipts	Receptionist, Manager	403 FORBIDDEN

1.4 Non-screen functions

Ten background jobs keep holds, bookings, plans and messages correct without anyone pressing a button. A single probe query checks every 15 seconds whether any job has work.

#	Function	Trigger	Description
1	Expire holds	Every 15 s, when due	Cancels court holds past their hold time and frees the court.
2	Mark no-shows	Every 15 s, when due	Marks a confirmed booking as a no-show 10 minutes after its start without check-in.
3	Complete bookings	Every 15 s, when due	Closes an in-use booking 10 minutes after it ends.
4	Send notifications	Every 15 s, when due	Sends up to 50 queued messages. A message is marked sent only on delivery and stops after 5 tries.
5	Withdraw payment links	Every 15 s, when due	Cancels a payOS link whose court hold has lapsed. Runs only when payOS is configured.
6	Expire waitlist offers	Every 15 s, when due	Expires an unanswered offer and offers the seat to the next member.
7	Lock attendance	Every 15 s, when due	Closes a session's register 2 hours after it ends.
8	Expire plans	Every 15 s, when due	Marks a plan expired the day after its end date.
9	Plan expiry reminders	Every 15 s, when due	Sends one in-app reminder 7 days, 3 days and 0 days before a plan ends.
10	Generate sessions	Every 10 minutes	Creates upcoming sessions for every open class.
11	payOS webhook	Called by payOS	Verifies the signature and amount, then posts the payment through the normal settle path.

The job loop starts with the first API request on each server instance. Each job with work runs in its own transaction, so one failure cannot roll back another.

1.5 Entity relationship diagram

The conceptual ERD shows 17 core entities in Chen notation, with a verb on each relationship and 1, N or M cardinality.

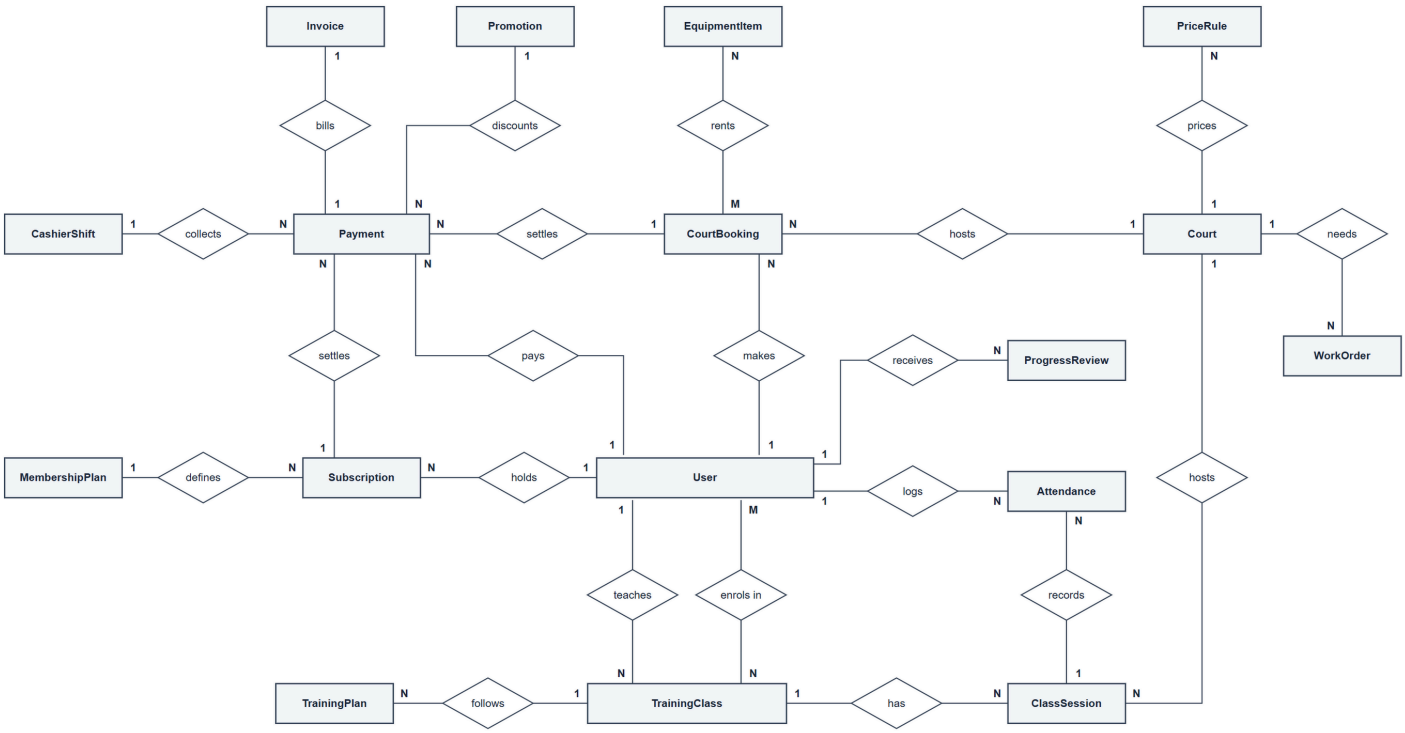


Figure 19. *Conceptual entity relationship diagram*

#	Entity	Table	Description
1	User	`users`	Every person with an account, with one role: member, receptionist, coach or manager.
2	Court	`courts`	A playable court for one sport, with a status of ready, maintenance or closed.
3	PriceRule	`price_rules`	Price per hour by sport, court class, day type and hour band.
4	CourtBooking	`court_bookings`	A court-time owned by a member or a walk-in, from hold to completed.
5	Payment	`payments`	Money in or out, with method, amount, VAT, shift and capture mode.
6	Invoice	`invoices`	The receipt issued for a payment, with a gapless code.
7	Promotion	`promotions`	A promo code with its discount, validity and usage limits.
8	CashierShift	`cashier_shifts`	One receptionist's till session, from opening float to counted close.
9	MembershipPlan	`membership_plans`	A plan on sale: price, duration, court hours or sessions, sport scope.
10	Subscription	`subscriptions`	A member's copy of a plan, with start and end dates and balances.
11	TrainingClass	`classes`	A coached class with a coach, a court, a weekly timetable and a capacity.
12	ClassSession	`sessions`	One dated meeting of a class.
13	Attendance	`attendance`	One member's mark for one session.
14	TrainingPlan	`training_plans`	Up to 12 timed blocks for a class, a session or one student.
15	ProgressReview	`progress_reviews`	An append-only review scoring technique, fitness and attitude.
16	EquipmentItem	`equipment_items`	Rentable gear with stock on hand.
17	WorkOrder	`work_orders`	A maintenance job that can block a court while it is open.

1.6 Class diagrams

Four class diagrams refine the ERD by subsystem, with attributes, operations, enumerations and multiplicities.

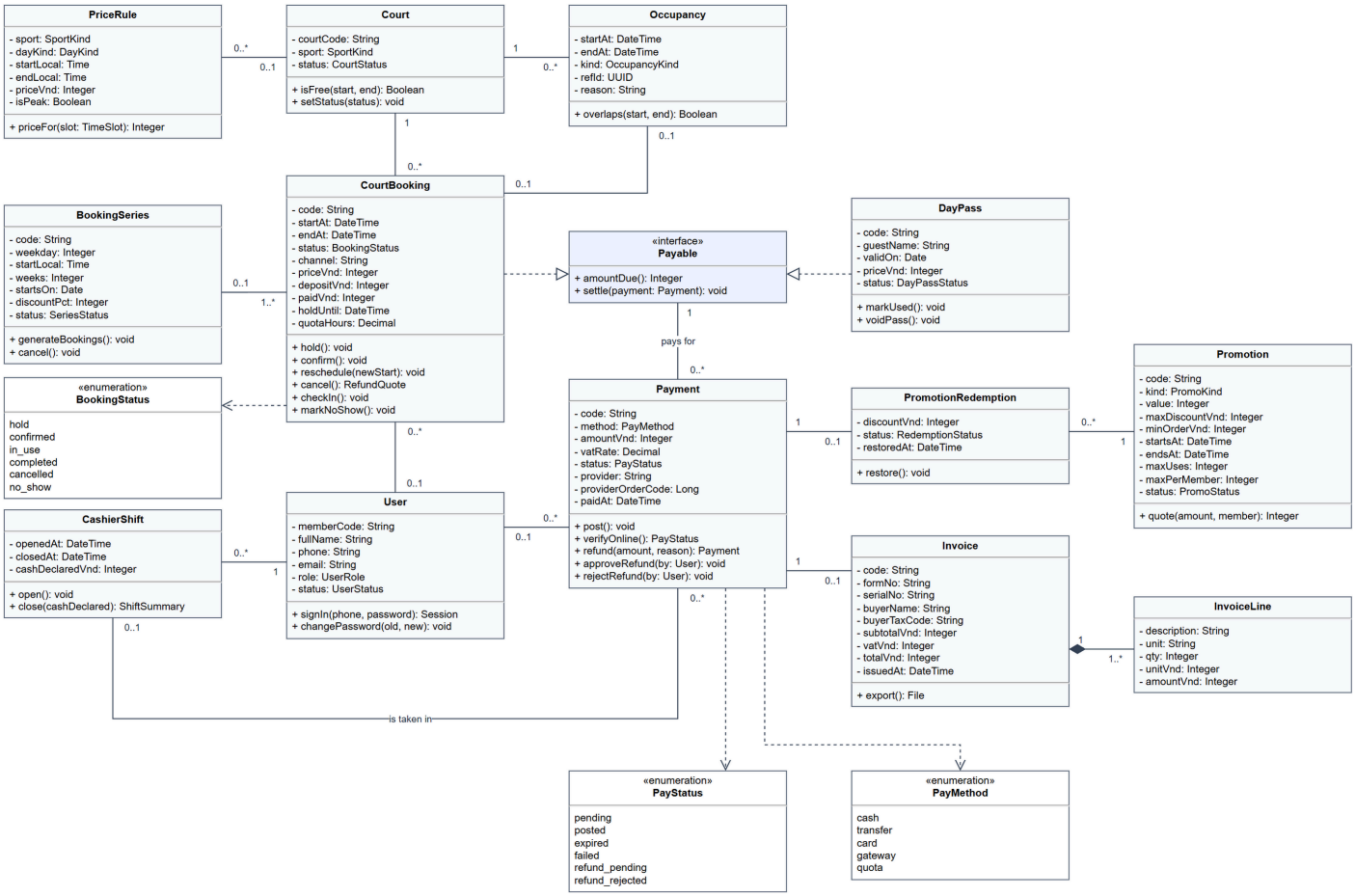


Figure 20. *Class diagram: booking and payment*

Figure 20 covers Court, PriceRule, Occupancy, CourtBooking, BookingSeries, DayPass, Payment, CashierShift, Promotion, PromotionRedemption, Invoice and InvoiceLine. CourtBooking, DayPass and Subscription realise the Payable interface.

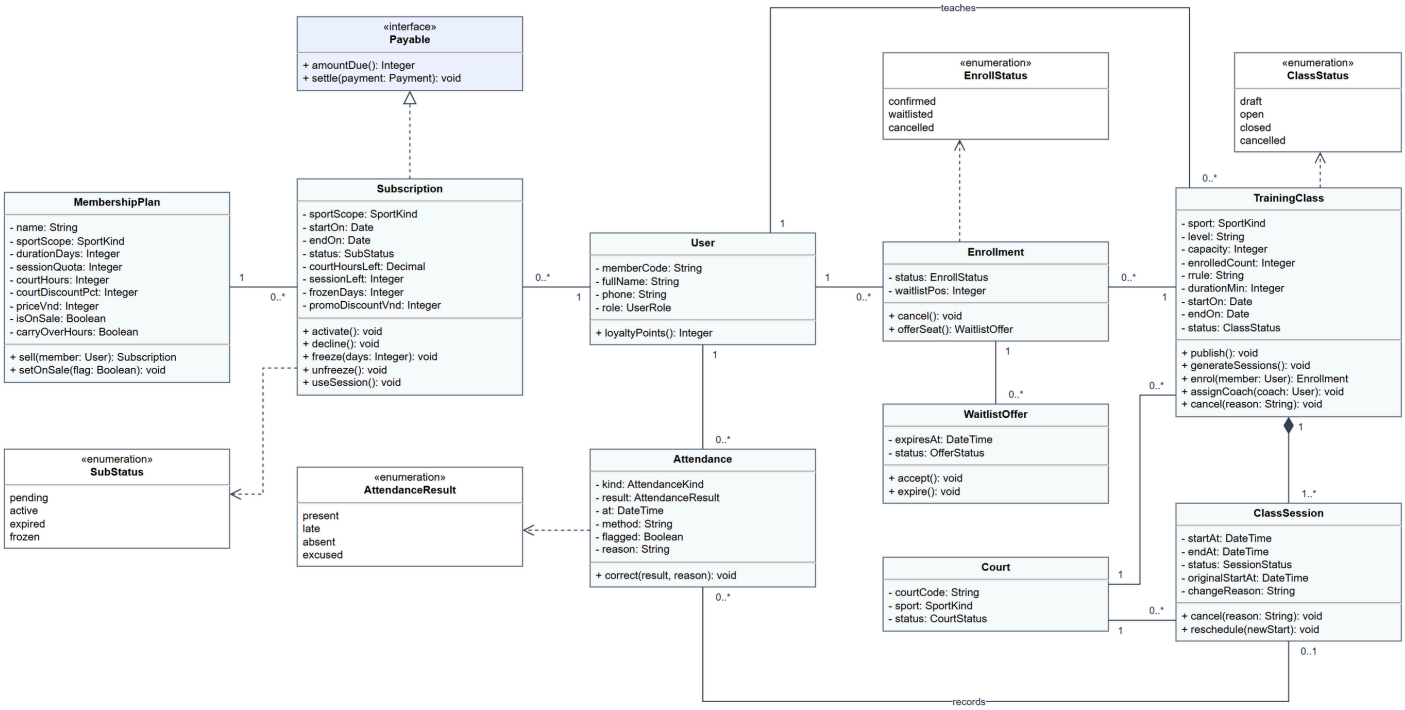


Figure 21. *Class diagram: membership and classes*

Figure 21 covers MembershipPlan, Subscription, TrainingClass, ClassSession, Enrollment, WaitlistOffer and Attendance, with their status enumerations.

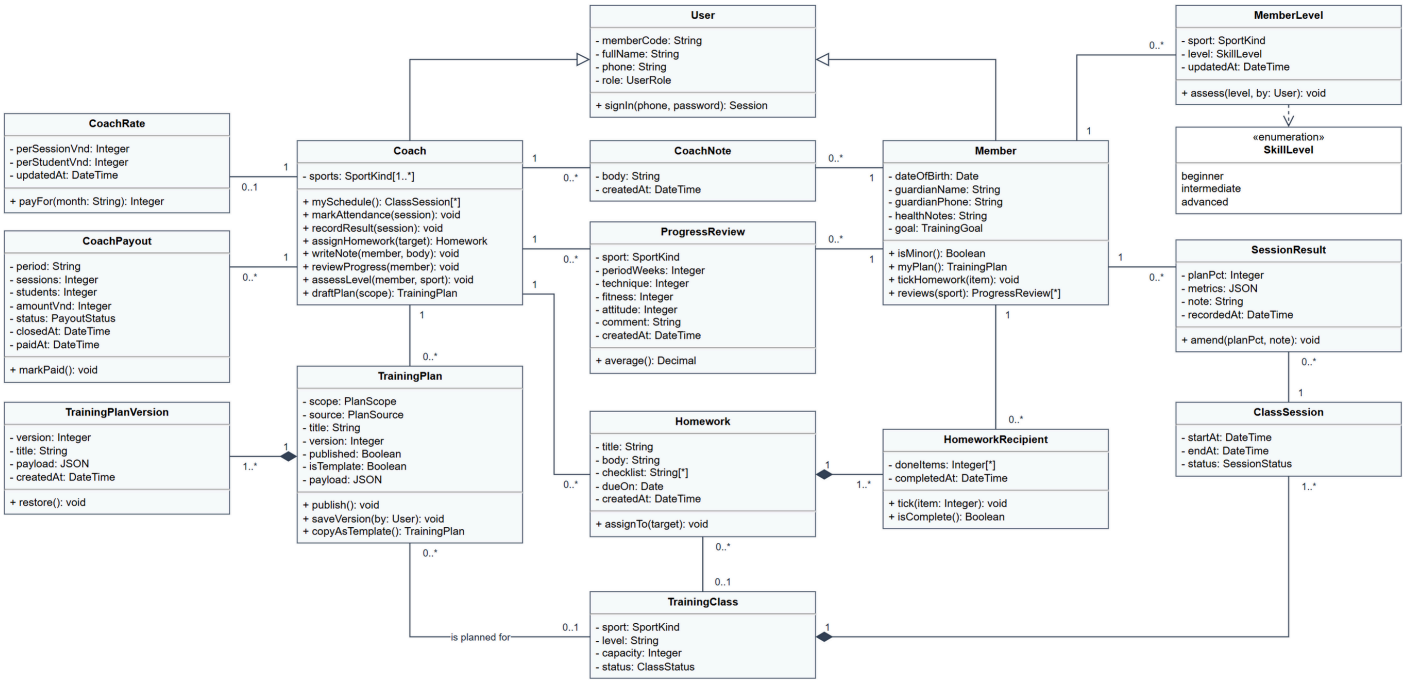


Figure 22. *Class diagram: coaching and training*

Figure 22 specialises User into Coach and Member. It covers CoachRate, CoachPayout, MemberLevel, SessionResult, TrainingPlan, TrainingPlanVersion, Homework, HomeworkRecipient, ProgressReview and CoachNote.

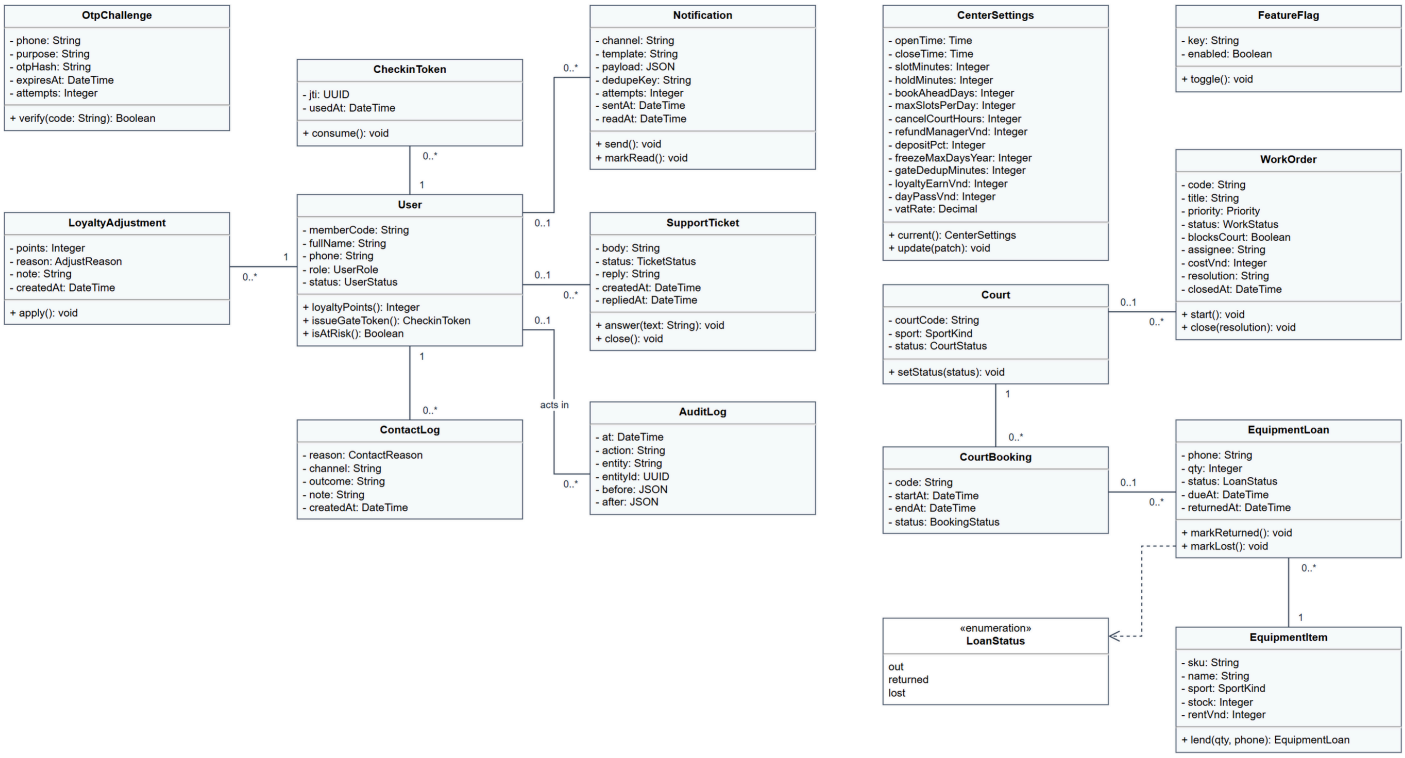


Figure 23. *Class diagram: centre operations*

Figure 23 covers CenterSettings, FeatureFlag, OtpChallenge, CheckinToken, Notification, SupportTicket, ContactLog, LoyaltyAdjustment, AuditLog, WorkOrder, EquipmentItem and EquipmentLoan.

2. Use case specifications

This section specifies all 44 use cases in the FPT template: a header table, then the normal flow, the alternative flows and the exceptions. Each alternative or exception is numbered by the step it branches from, so 3a branches from step 3.

Every refusal named below reaches the screen as a sentence, with its error code and, where a business rule applies, the rule code from V.1. The common requirements in V.3 (idempotency keys, audit rows, sessions and rate limits) apply to every use case and are not repeated.

2.1 UC-01 Browse courts, classes and prices

Field	Detail
UC ID and name	UC-01 Browse courts, classes and prices
Created by / date	Arena3 team / 2026-10-12
Primary actor	Guest
Secondary actors	None
Trigger	The guest opens the public site (/).
Description	A guest reads the sports, courts, opening hours, plan prices and open classes before deciding to join.
Preconditions	None.
Postconditions	No data changes.
Priority	Must have
Frequency of use	Many times a day
Business rules	BR-38

Normal flow

1. The guest opens the landing page.
2. The system shows the sports, the opening hours, the court list and the plan prices.
3. The guest opens the class list.
4. The system shows each open class with its sport, coach, schedule and seats left.
5. The guest selects Create an account to continue (UC-02).

Alternative flows

- 2a. The guest switches the language: the page is shown again in English or Vietnamese.

Exceptions

- 4a. No class is open: the system says so and shows the plan prices only.

2.2 UC-02 Register a member account

Field	Detail
UC ID and name	UC-02 Register a member account
Created by / date	Arena3 team / 2026-10-12
Primary actor	Guest
Secondary actors	Email service (SMTP)
Trigger	The guest selects Create an account on /register.
Description	A guest creates a member account and proves the email address with a one-time code.
Preconditions	The guest is signed out. The phone and email have no account yet.
Postconditions	A member account exists, the guest is signed in for seven days, and an audit row is written.
Priority	Must have
Frequency of use	Once per new member
Business rules	BR-01, BR-02, BR-03, BR-04

Normal flow

1. The guest enters full name, phone, email, password and date of birth.
2. The guest accepts the terms and the data-privacy notice, then submits.
3. The system checks every field and confirms the phone and email are unused.
4. The system stores a sign-up challenge for five minutes and emails a one-time code.
5. The guest enters the code on the same page.
6. The system checks the code, creates the member account and returns a session token.
7. The system opens My schedule.

Alternative flows

- 1a. The guest is under 16: the guest also enters a guardian's name and phone (BR-03).

Exceptions

- 3a. A field is malformed: VALIDATION, and the field to correct is marked.
- 3b. The phone or email already has an account: BR-01.
- 3c. The password is shorter than 8 characters or lacks letters and digits: BR-02.
- 3d. The terms are not accepted: BR-04.
- 6a. The code is wrong: the attempt is counted and the guest may try again.
- 6b. The code is older than five minutes: the guest submits the form again to receive a new code.
- 6c. Five wrong attempts: the challenge is locked for 15 minutes (RATE_LIMITED).

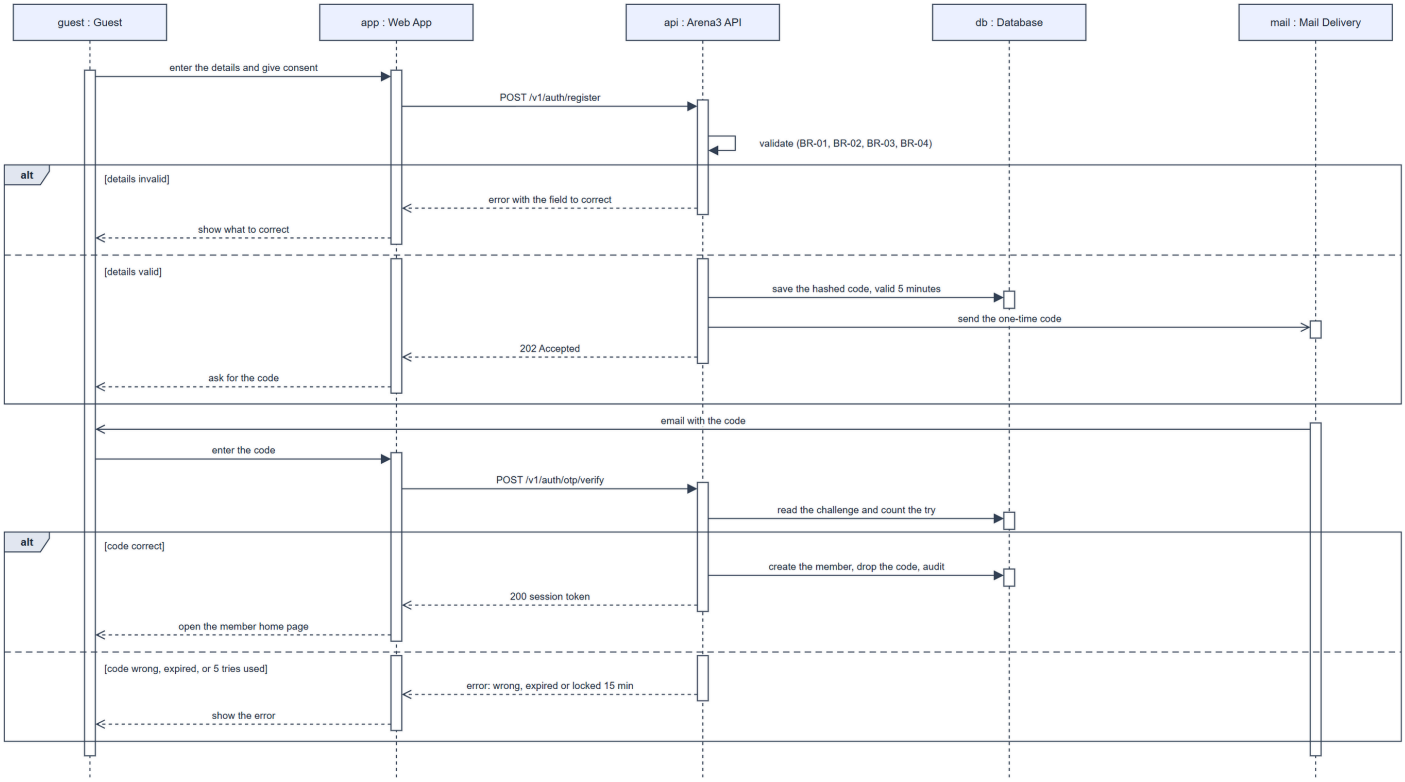


Figure 24. *Sequence diagram: register a member account*

The activity diagram in Figure 2 (I.3.1) shows the same flow.

2.3 UC-03 Sign in and sign out

Field	Detail
UC ID and name	UC-03 Sign in and sign out
Created by / date	Arena3 team / 2026-10-12
Primary actor	Registered User
Secondary actors	None
Trigger	The user selects Sign in on /login, or Sign out in any portal.
Description	A registered user opens a session with phone or email and password, and closes it when done.
Preconditions	The user has an active account.
Postconditions	Sign in: a session token is stored as a hash and the user lands on the home screen of their role. Sign out: the session is revoked.
Priority	Must have
Frequency of use	Many times a day
Business rules	None

Normal flow

1. The user enters a phone or email and the password.
2. The system checks the password against the stored salted hash.
3. The system creates a session: seven days for a member, twelve hours for staff.
4. The system opens the portal for the user's role: /app, /desk, /coach or /manager.
5. Later, the user selects Sign out.
6. The system revokes the session and opens the landing page.

Alternative flows

- 4a. The user opened a protected page before signing in: the system returns the user to that page.

Exceptions

- 2a. The password is wrong: UNAUTHENTICATED, and the attempt is counted.
- 2b. Five wrong passwords: sign-in is locked for 15 minutes (RATE_LIMITED).
- 2c. The account is locked or disabled: FORBIDDEN, and the user is told to contact the centre.

2.4 UC-04 Reset a forgotten password

Field	Detail
UC ID and name	UC-04 Reset a forgotten password
Created by / date	Arena3 team / 2026-10-12
Primary actor	Registered User
Secondary actors	Email service (SMTP)
Trigger	The user selects Forgot password on /login.

Field	Detail
Description	A user who has forgotten the password proves control of the account email with a one-time code and sets a new password.
Preconditions	The user is signed out.
Postconditions	The password is replaced, every other session of the user is revoked, and an audit row is written.
Priority	Must have
Frequency of use	A few times a week
Business rules	BR-02

Normal flow

1. The user enters the phone number of the account on /forgot.
2. The system emails a one-time code valid for five minutes to the address on the account.
3. The user enters the code and a new password twice.
4. The system checks the code and the password rules.
5. The system stores the new password and signs the user in.

Exceptions

- 2a. The phone has no account: the system shows the same message as for a real account, so it never reveals who is a member.
- 4a. The code is wrong or expired: the user may try again or request a new code.
- 4b. Five wrong codes: the reset is locked for 15 minutes (RATE_LIMITED).
- 4c. The new password breaks BR-02, or the two entries differ: VALIDATION.

2.5 UC-05 Manage the personal profile

Field	Detail
UC ID and name	UC-05 Manage the personal profile
Created by / date	Arena3 team / 2026-10-12
Primary actor	Registered User
Secondary actors	None
Trigger	The user opens Account (/account).
Description	A signed-in user updates the profile details and changes the password.
Preconditions	The user is signed in.
Postconditions	The profile or password is updated and an audit row is written.
Priority	Should have
Frequency of use	Occasionally
Business rules	BR-01, BR-02

Normal flow

1. The user opens Account.

2. The system shows the name, phone, email, date of birth and language.
3. The user edits the details and saves.
4. The system checks the fields and stores them.

Alternative flows

- 3a. The user changes the password: the user enters the current password and the new one twice. The system checks the current password, rejects reuse of the same password and stores the new hash.

Exceptions

- 3b. The current password is wrong: VALIDATION.
- 3c. The new password breaks BR-02 or the two entries differ: VALIDATION.
- 4a. The new phone or email belongs to another account: BR-01.

2.6 UC-06 View notifications

Field	Detail
UC ID and name	UC-06 View notifications
Created by / date	Arena3 team / 2026-10-12
Primary actor	Registered User
Secondary actors	None
Trigger	The user opens Notifications (/app/notifications or /alerts), or a new badge appears.
Description	The user reads booking, payment, class, plan and support notices, and marks them as read.
Preconditions	The user is signed in.
Postconditions	The notices shown are marked as read.
Priority	Should have
Frequency of use	Daily
Business rules	None
Other information	Notices are queued and sent in batches of 50, with up to five attempts each. Plan reminders go 7, 3 and 0 days before the end date.

Normal flow

1. The user opens Notifications.
2. The system lists the user's notices, newest first, with unread ones highlighted.
3. The user opens a notice; the system marks it as read and links to the related screen.

Alternative flows

- 3a. The user selects Mark all as read: every notice is marked as read.

Exceptions

- 2a. The user has no notices: the system shows an empty list.

2.7 UC-07 View my schedule

Field	Detail
UC ID and name	UC-07 View my schedule
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	None
Trigger	The member opens My schedule (/app).
Description	The member sees upcoming court bookings, class sessions, plan balance and recent attendance in one place.
Preconditions	The member is signed in.
Postconditions	No data changes.
Priority	Must have
Frequency of use	Daily
Business rules	BR-12

Normal flow

1. The member opens My schedule.
2. The system lists upcoming bookings and class sessions in time order.
3. The system shows the active plan with its end date, sessions left and court hours left.
4. The system shows the attendance recorded for recent sessions.
5. The member opens a booking to reschedule or cancel it (UC-09).

Alternative flows

- 3a. The plan ends within seven days: the system shows a renewal notice.

Exceptions

- 2a. The member has nothing booked: the system suggests Book a court and Classes.

2.8 UC-08 Book a court and pay

Field	Detail
UC ID and name	UC-08 Book a court and pay
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	payOS
Trigger	The member selects a free slot on Book a court (/app/book).
Description	The member holds a court slot for five minutes and pays the amount due to confirm it.
Preconditions	The member is signed in. The slot is in the future, within seven days and inside opening hours. The court is ready.
Postconditions	Success: the booking is confirmed and the payment is posted with a receipt code. Failure: the hold lapses and the slot is free again.
Priority	Must have

Field	Detail
Frequency of use	Many times a day
Business rules	BR-05, BR-06, BR-07, BR-08, BR-09, BR-10, BR-13, BR-37, BR-39, BR-40, BR-41, BR-42

Normal flow

1. The member chooses a sport and a date, then a free slot on the court grid.
2. The system checks the booking window, the court status and the daily cap of two slots.
3. The system prices the slot from the price rules and holds it for five minutes.
4. The member may enter a promo code; the system applies it to the amount due.
5. The member chooses Pay online. The system creates a payOS link and shows the VietQR code.
6. The member scans the QR with a banking app and pays.
7. payOS confirms by a signed webhook, or the screen's check every three seconds reads the link as paid.
8. The system compares the amount with the amount due, posts the payment and confirms the booking.
9. The system shows the receipt, and the booking appears in My schedule.

Alternative flows

- 2a. The slot clashes with one of the member's classes: the system asks for confirmation (BR-09), and the member may hold it anyway.
- 3a. The slot is at peak time: the member pays a 30% deposit now, and the desk collects the balance on arrival (UC-22).
- 3b. The member already holds a slot: the new hold is checked in full before it replaces the old one (BR-10).
- 5a. The member pays from plan court hours: the system deducts the hours and confirms at once (BR-13).
- 5b. The member pays by bank transfer: the booking waits up to 120 minutes until the desk confirms the transfer (UC-22).

Exceptions

- 2b. The slot is in the past (BR-07), more than seven days ahead (BR-05), or outside opening hours (BR-06).
- 2c. The court is not ready (BR-06), or the member already holds two slots that day (BR-05).
- 3c. No price rule matches the slot: BR-37; the system never invents a price.
- 3d. Another member took the slot first: CONFLICT_SLOT.
- 4a. The promo code is unknown, expired or out of scope (BR-40), or used up (BR-41). A code that does not stack gives way to a larger plan discount (BR-42).
- 5c. The plan has too few court hours left: BR-13.
- 5d. Online payment is switched off or cannot start: C-08 or C-09, and the member pays at the desk.

- 8a. The hold lapsed before payment was confirmed: HOLD_EXPIRED, and the member books again.

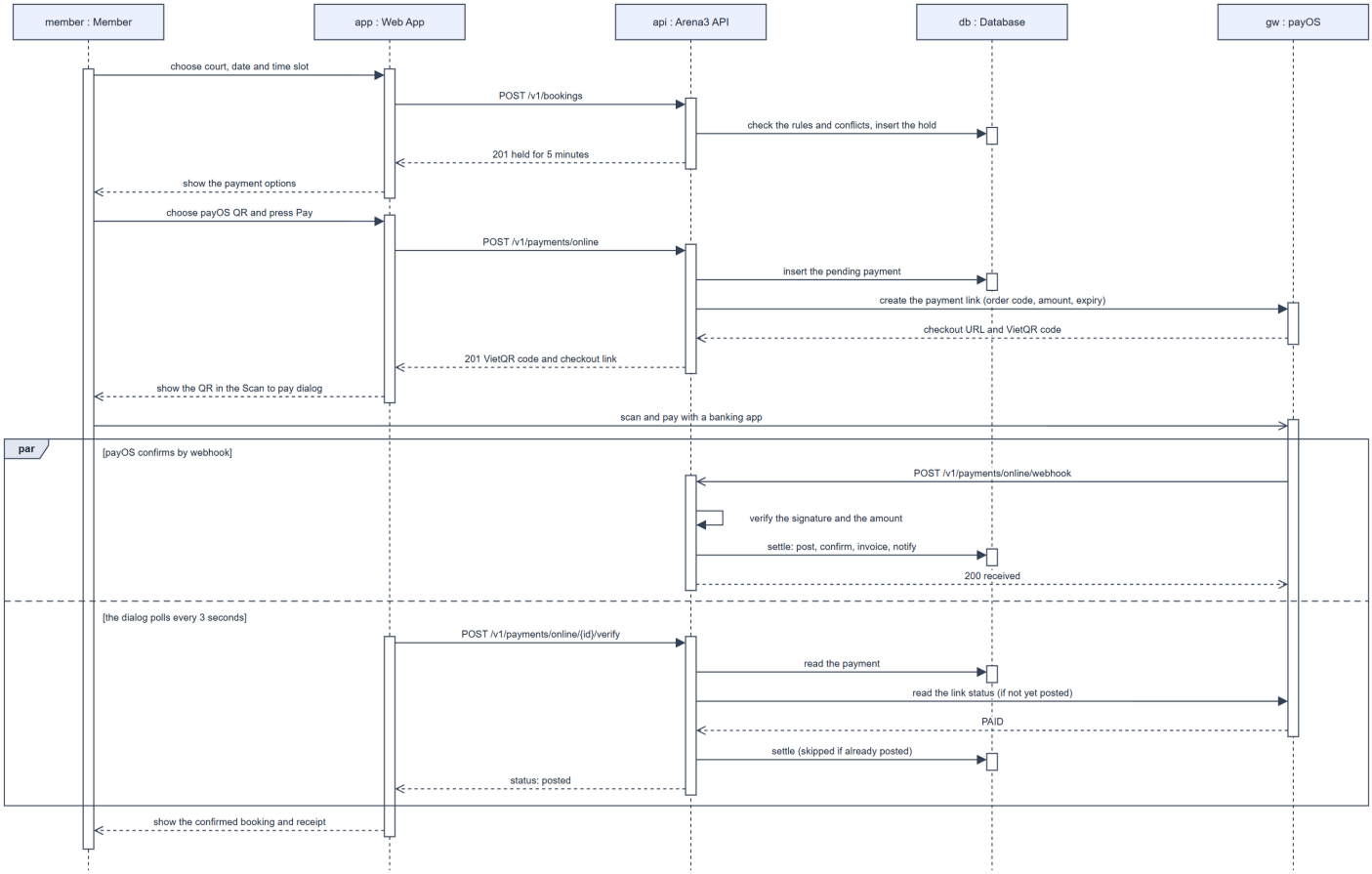


Figure 25. *Sequence diagram: book a court and pay by payOS QR*

The activity diagram in Figure 3 (I.3.2) shows the three payment paths.

2.9 UC-09 Reschedule or cancel a booking

Field	Detail
UC ID and name	UC-09 Reschedule or cancel a booking
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	None
Trigger	The member opens a booking in My schedule and selects Reschedule or Cancel.
Description	The member moves a confirmed booking to another slot, or cancels it and receives a refund by the notice given.
Preconditions	The member owns the booking. The booking is confirmed and has not started.
Postconditions	Reschedule: the booking owns the new slot and the old slot is free. Cancel: the booking is cancelled, the slot is free, and any refund or plan hour is returned.
Priority	Must have
Frequency of use	Several times a day
Business rules	BR-05, BR-06, BR-07, BR-09

Normal flow

1. The member opens the booking and selects Reschedule.
2. The member picks a new slot for the same sport.
3. The system checks the new slot with the booking rules and that its price equals the price paid.
4. The system moves the booking and frees the old slot.

Alternative flows

- 1a. The member selects Cancel instead: the system quotes the refund by notice (100% at 24 hours or more, 50% at 4 to 24 hours, 0% under 4 hours).
- 1b. The member confirms: the system cancels the booking, frees the slot and raises the refund. Plan hours come back when the refund is 50% or more.
- 1c. The booking is still an unpaid hold: it is released at no cost.

Exceptions

- 1d. The booking is in a state that cannot be cancelled or moved: CONFLICT_STATE.
- 1e. The booking starts in less than 2 hours: it can no longer be moved.
- 3a. The new slot breaks a booking rule (BR-05, BR-06, BR-07) or clashes with a class (BR-09).
- 3b. The new slot has a different price: the member cancels and books the new slot instead.
- 3c. Another member took the new slot first: CONFLICT_SLOT.

2.10 UC-10 Order a membership plan

Field	Detail
UC ID and name	UC-10 Order a membership plan

Field	Detail
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	Receptionist, who takes payment; payOS
Trigger	The member selects Order on a plan in Plans (/app/plans).
Description	The member orders a membership plan online and pays for it at the front desk.
Preconditions	The member is signed in. The plan is on sale.
Postconditions	A plan order is pending payment. Once paid, the plan becomes active from its start date.
Priority	Must have
Frequency of use	Daily
Business rules	BR-14, BR-16, BR-40

Normal flow

1. The member opens Plans and compares the plans on sale.
2. The member selects a plan and may enter a promo code.
3. The system checks the plan and the code and creates a pending order.
4. The system tells the member to pay at the front desk.
5. The receptionist takes payment for the order (UC-20), and the plan becomes active.

Alternative flows

- 5a. The receptionist raises a payOS link for the order, and the member pays by VietQR.

Exceptions

- 3a. The plan has been withdrawn: BR-16.
- 3b. The member's current plan is frozen: BR-14.
- 3c. The promo code does not apply: BR-40.
- 3d. The member tries to pay for a plan online: the system answers that the front desk takes payment for plans.
- 3e. More than 3 orders a minute or 10 an hour: RATE_LIMITED.

2.11 UC-11 View receipts and redeem points

Field	Detail
UC ID and name	UC-11 View receipts and redeem points
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	None
Trigger	The member opens Points (/app/points) or a receipt from My schedule.
Description	The member views paid receipts and e-invoices, and turns loyalty points into a single-use discount code.
Preconditions	The member is signed in.

Field	Detail
Postconditions	Redeem: points are spent and a promo code worth their value is created.
Priority	Should have
Frequency of use	Weekly
Business rules	BR-40, BR-41
Other information	One point is earned for every 10,000đ paid for bookings, plans and day passes. Spending points never lowers the tier.

Normal flow

1. The member opens Points.
2. The system shows the points balance, the tier and the points needed for the next tier.
3. The member enters the points to redeem.
4. The system creates a single-use code worth 100đ per point, valid for 60 days, and lowers the balance.
5. The member uses the code at checkout (UC-08 or UC-10).

Alternative flows

- 1a. The member opens a receipt instead: the system shows the payment and its e-invoice as PDF.

Exceptions

- 4a. Fewer than 50 points are entered, or more than the balance: VALIDATION.

2.12 UC-12 Check in by self-service

Field	Detail
UC ID and name	UC-12 Check in by self-service
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	None
Trigger	The member opens Gate pass (/app/pass) at the centre.
Description	The member shows a short-lived QR pass at the desk, or scans the desk code to check in without staff.
Preconditions	The member is signed in.
Postconditions	Self check-in: one visit row exists for the member.
Priority	Should have
Frequency of use	Daily
Business rules	BR-30, BR-32

Normal flow

1. The member opens Gate pass.
2. The system shows a short-lived member QR code that refreshes itself.
3. The member shows the code to the receptionist, who continues with UC-26.

Alternative flows

- 3a. Self check-in is switched on: the member scans the desk code with the phone, and the system records the visit as in UC-26.

Exceptions

- 3b. Self check-in is switched off: BR-30, and the member shows the pass at the desk.
- 3c. The scanned code is not the desk code, or was already used: BR-30.

2.13 UC-13 Enrol in a class

Field	Detail
UC ID and name	UC-13 Enrol in a class
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	Receptionist or manager, enrolling on the member's behalf
Trigger	The member selects Enrol on a class in Classes (/app/classes).
Description	The member takes a seat in an open class, or joins its waitlist when the class is full.
Preconditions	The member is signed in and holds an active plan covering the class sport.
Postconditions	The member is enrolled and the sessions appear in My schedule, or the member is waitlisted with a position.
Priority	Must have
Frequency of use	Daily
Business rules	BR-12, BR-18, BR-19, BR-20, BR-21, BR-22

Normal flow

1. The member opens Classes and chooses an open class.
2. The system checks that a plan covers the sport and has sessions left.
3. The system checks the class does not clash with the member's other classes.
4. A seat is free: the system enrols the member and adds the class sessions to My schedule.

Alternative flows

- 4a. The class is full: the system adds the member to the waitlist and shows the position.
- 4b. A seat frees up later: the next member in line receives an offer valid for two hours (UC-14).

Exceptions

- 1a. The class is not open for enrolment: BR-18.
- 2a. No active plan covers the sport: BR-19.
- 2b. The plan has no sessions left: BR-20.
- 3a. The class clashes with another class, or the member is already enrolled: BR-22.

The activity diagram in Figure 4 (I.3.3) shows this flow with its waitlist branch.

2.14 UC-14 Accept a waitlist offer

Field	Detail
UC ID and name	UC-14 Accept a waitlist offer
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	None
Trigger	The member receives a waitlist offer notice.
Description	A waitlisted member accepts a freed seat before the offer expires.
Preconditions	The member holds an offer that has not expired.
Postconditions	The member is enrolled and the offer is closed.
Priority	Should have
Frequency of use	A few times a week
Business rules	BR-19, BR-21, BR-23

Normal flow

1. The member opens the offer from Notifications or Classes.
2. The member selects Accept.
3. The system checks the offer, the seat and the member's plan.
4. The system enrols the member and closes the offer.

Alternative flows

- 2a. The member ignores the offer: after two hours the scheduled job expires it and offers the seat to the next member (BR-23).

Exceptions

- 3a. The offer has expired: BR-23.
- 3b. The seat was filled in the meantime: BR-21.
- 3c. The plan no longer covers the sport: BR-19.

2.15 UC-15 Leave a class

Field	Detail
UC ID and name	UC-15 Leave a class
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	None
Trigger	The member selects Leave on an enrolled class.
Description	The member leaves a class in time, which frees the seat for the waitlist.
Preconditions	The member is enrolled in the class.
Postconditions	The enrolment is cancelled, and the next waitlisted member receives an offer.
Priority	Should have

Field	Detail
Frequency of use	A few times a week
Business rules	BR-24

Normal flow

1. The member opens the class in My schedule and selects Leave.
2. The system checks the time to the next session.
3. The system cancels the enrolment and removes the future sessions from My schedule.
4. The system offers the seat to the first member on the waitlist.

Exceptions

- 2a. The next session starts in less than 4 hours: BR-24, and the member stays enrolled.

2.16 UC-16 Follow my training progress

Field	Detail
UC ID and name	UC-16 Follow my training progress
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	Coach, who writes the plans and feedback
Trigger	The member opens Training (/app/train).
Description	The member follows training plans, coach feedback, reviews and homework, and sets a personal goal.
Preconditions	The member is signed in and enrolled in at least one class. The training module is switched on.
Postconditions	Homework progress and the goal are saved.
Priority	Could have
Frequency of use	Weekly
Business rules	BR-47, BR-52

Normal flow

1. The member opens Training.
2. The system shows the plan of each upcoming session, the latest session results and the coach's reviews.
3. The member opens a homework item and ticks the checklist items done.
4. The system saves the progress, which the coach sees on the student profile.
5. The member sets or edits a personal goal; the system saves it.

Exceptions

- 1a. The training module is switched off: FORBIDDEN with BR-52.
- 3a. The homework is past its due date: it is shown as overdue but can still be ticked.

2.17 UC-17 Get help from the AI assistant or staff

Field	Detail
UC ID and name	UC-17 Get help from the AI assistant or staff
Created by / date	Arena3 team / 2026-10-12
Primary actor	Member
Secondary actors	Gemini AI service; Receptionist, who answers support requests
Trigger	The member opens Assistant (/app/assistant).
Description	The member asks the AI assistant about the centre, or sends a support request to the front desk.
Preconditions	The member is signed in.
Postconditions	An answer is shown, or a support request is open in the desk queue.
Priority	Could have
Frequency of use	Daily
Business rules	BR-46

Normal flow

1. The member types a question of up to 800 characters.
2. The system sends the question with the centre facts to the Gemini model, without health notes.
3. The system shows the answer.

Alternative flows

- 1a. The member writes a support request instead: the system opens a ticket for the desk (UC-29), and the reply arrives as a notice.

Exceptions

- 2a. More than 20 questions in a minute: RATE_LIMITED.
- 2b. The model is unavailable: the system answers from the centre facts and suggests a support request.

2.18 UC-18 Open and close a cashier shift

Field	Detail
UC ID and name	UC-18 Open and close a cashier shift
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	None
Trigger	The receptionist starts or ends a working shift on Payments (/desk/payments).
Description	The receptionist opens a cashier shift with a float and closes it by declaring the cash counted.
Preconditions	The receptionist is signed in.
Postconditions	Open: the shift is open and payments are tied to it. Close: the shift is closed with the expected and declared cash and the difference.

Field	Detail
Priority	Must have
Frequency of use	Twice a day per receptionist
Business rules	BR-44

Normal flow

1. The receptionist selects Open shift and enters the opening float.
2. The system opens the shift.
3. During the shift, every payment taken at the desk is tied to it.
4. At the end, the receptionist selects Close shift and enters the cash counted.
5. The system closes the shift and shows expected cash, declared cash and the difference.

Exceptions

- 1a. The receptionist already has an open shift: CONFLICT_STATE.
- 4a. The counted cash is missing: VALIDATION.

2.19 UC-19 Manage a member record at the desk

Field	Detail
UC ID and name	UC-19 Manage a member record at the desk
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Coach (read only, own students)
Trigger	The receptionist searches for a member on the desk home (/desk) or opens Member profile (/desk/member/$id).
Description	The receptionist registers a walk-in customer, views and edits a member record, and follows up members at risk of leaving.
Preconditions	The receptionist is signed in.
Postconditions	A member record is created or updated, or a follow-up contact is logged. An audit row is written.
Priority	Must have
Frequency of use	Many times a day
Business rules	BR-01, BR-03, BR-04, BR-36, BR-46

Normal flow

1. The receptionist searches by name, phone or member code.
2. The system shows the matching members.
3. The receptionist opens a member profile.
4. The system shows details, plans, bookings, payments, visits and attendance.
5. The receptionist edits the details and saves; the system checks and stores them.

Alternative flows

- 1a. The person has no account: the receptionist registers them at the counter with name, phone, terms consent and, for a minor, guardian details.
- 1b. The receptionist opens At-risk members (/desk/at-risk): the system lists members with a live plan but no visit or class for the idle period, or with three absences in a row, and the receptionist logs a follow-up contact.
- 5a. The receptionist resets the member's password: the system issues a temporary password.

Exceptions

- 1c. The terms are not accepted: BR-04.
- 5b. The phone already has an account: BR-01.
- 5c. A minor has no guardian details: BR-03.

2.20 UC-20 Manage a member plan at the desk

Field	Detail
UC ID and name	UC-20 Manage a member plan at the desk
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Member
Trigger	The receptionist opens the Plans panel on Member profile.
Description	The receptionist sells, renews, freezes and resumes a member's plan.
Preconditions	The receptionist is signed in with an open shift. The member exists.
Postconditions	Sell or renew: a posted payment, an e-invoice and an active plan. Freeze or resume: the plan status and end date change. An audit row is written.
Priority	Must have
Frequency of use	Many times a day
Business rules	BR-12, BR-14, BR-15, BR-16, BR-39, BR-40, BR-44

Normal flow

1. The receptionist selects a plan, or a pending order the member placed (UC-10).
2. The system prices it, applies any promo code and shows the amount due.
3. The receptionist takes cash, card or transfer and confirms.
4. The system posts the payment, issues the e-invoice and activates the plan.

Alternative flows

- 1a. Renew: the new plan starts the day after the current one ends.
- 1b. Freeze: the receptionist enters 1 to 90 days; the system freezes the plan and moves the end date by that many days.
- 1c. Resume: the system sets a frozen plan active again.
- 3a. The member pays by VietQR: the receptionist raises a payOS link for the order.

Exceptions

- 1d. The plan has been withdrawn: BR-16.
- 1e. Freeze on a plan that is not active: BR-14. Over the yearly allowance: BR-15.
- 1f. Resume on a plan that is not frozen: CONFLICT_STATE.
- 3b. No open shift: BR-44.

2.21 UC-21 Book a court at the desk

Field	Detail
UC ID and name	UC-21 Book a court at the desk
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Member, if known
Trigger	The receptionist selects a slot on Courts (/desk/courts) or opens Fixed series (/desk/series).
Description	The receptionist books a court for a walk-in customer, or a fixed weekly series of 4 to 12 weeks at 10% off.
Preconditions	The receptionist is signed in with an open shift.
Postconditions	Walk-in: a confirmed, paid booking. Series: one booking per week, all confirmed, with one payment.
Priority	Must have
Frequency of use	Many times a day
Business rules	BR-05, BR-06, BR-07, BR-08, BR-37, BR-44

Normal flow

1. The receptionist selects a free slot on the court map.
2. The receptionist enters the customer's name and phone and selects Walk-in.
3. The system checks the slot and prices it.
4. The receptionist takes payment.
5. The system confirms the booking and posts the payment.

Alternative flows

- 1a. Fixed series: the receptionist picks the court, weekday, time and 4 to 12 weeks. The system previews every week, marks any week that is taken, and books the free weeks at 10% off after payment.

Exceptions

- 2a. The name, phone or court is missing: VALIDATION.
- 3a. The slot has less than 20 minutes left: BR-07.
- 3b. The slot is outside opening hours or the court is not ready: BR-06.
- 3c. No price rule matches: BR-37.
- 3d. Another booking took the slot first: CONFLICT_SLOT.
- 4a. No open shift: BR-44.

2.22 UC-22 Settle pending payments

Field	Detail
UC ID and name	UC-22 Settle pending payments
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Member; payOS
Trigger	The receptionist opens Payments (/desk/payments) or a booking on Courts.
Description	The receptionist settles money that is still open: a booking balance, a bank transfer to confirm, or a payment for an order.
Preconditions	The receptionist is signed in with an open shift.
Postconditions	The payment is posted with a receipt and e-invoice, and the booking or order it pays for is confirmed.
Priority	Must have
Frequency of use	Many times a day
Business rules	BR-39, BR-44

Normal flow

1. The receptionist opens the list of pending items.
2. The receptionist selects a booking with a balance left after its deposit.
3. The receptionist takes the balance in cash, card or transfer.
4. The system posts the payment and marks the booking as paid.

Alternative flows

- 2a. A bank transfer is waiting: the receptionist checks the bank statement and selects Confirm; the system confirms the booking.
- 2b. The transfer has not arrived: the receptionist selects Reject; the system releases the slot and notifies the member.
- 2c. An order has no money taken yet: the receptionist may cancel it.

Exceptions

- 2d. The booking is not confirmed or has nothing left to pay: CONFLICT_STATE.
- 2e. The transfer is not confirmed within 120 minutes: the scheduled job releases the slot.
- 3a. No open shift: BR-44.

2.23 UC-23 Sell and redeem a day pass

Field	Detail
UC ID and name	UC-23 Sell and redeem a day pass
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Guest or member

Field	Detail
Trigger	The receptionist opens Day passes (/desk/day-passes).
Description	The receptionist sells a day pass for entry on a chosen day and marks it used when the customer arrives.
Preconditions	The receptionist is signed in with an open shift.
Postconditions	Sell: an issued pass with a posted payment and e-invoice. Redeem: the pass is used.
Priority	Should have
Frequency of use	Daily
Business rules	BR-44

Normal flow

1. The receptionist enters the customer's name, or the phone of a member, and the day.
2. The system prices the pass at the centre day-pass price (60,000đ by default).
3. The receptionist takes payment.
4. The system issues the pass with its own code, posts the payment and issues the e-invoice.
5. On the day, the receptionist finds the pass and selects Use.

Alternative flows

- 5a. The pass was sold in error and not used: the receptionist voids it and the payment is refunded.

Exceptions

- 1a. The day is in the past: VALIDATION.
- 3a. No open shift: BR-44.
- 5b. The pass is used on another day, or was already used or voided: CONFLICT_STATE.

2.24 UC-24 Raise a refund

Field	Detail
UC ID and name	UC-24 Raise a refund
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Manager, who signs off large refunds; Member, who is notified
Trigger	The receptionist chooses Raise a refund on a payment in Member profile.
Description	Staff give back all or part of a posted payment, and large refunds wait for a second person.
Preconditions	The receptionist is signed in. The payment is posted and has money left to refund.
Postconditions	A negative payment row exists with its own receipt code, either posted or pending approval. An audit row is written.
Priority	Must have
Frequency of use	A few times a week
Business rules	BR-44, BR-53

Normal flow

1. The receptionist opens the member profile, selects a posted payment and chooses Raise a refund.
2. The receptionist enters the amount and a reason.
3. The system works out what is still refundable: the payment, less refunds already posted or pending.
4. The amount is under the refund limit (1,000,000đ): the system posts the refund with its own receipt code.
5. The system writes an audit row and shows the refund receipt.
6. If the order is now refunded in full, a used promo code is given back to the member.

Alternative flows

- 4a. The amount is at or above the refund limit: the refund is saved as pending approval and goes to a manager (UC-42).
- 4b. A manager raises the refund: it is posted directly, whatever the amount.

Exceptions

- 3a. The payment is already refunded in full, or the amount is larger than what remains: VALIDATION.
- 4c. A cash refund without an open shift: BR-44.

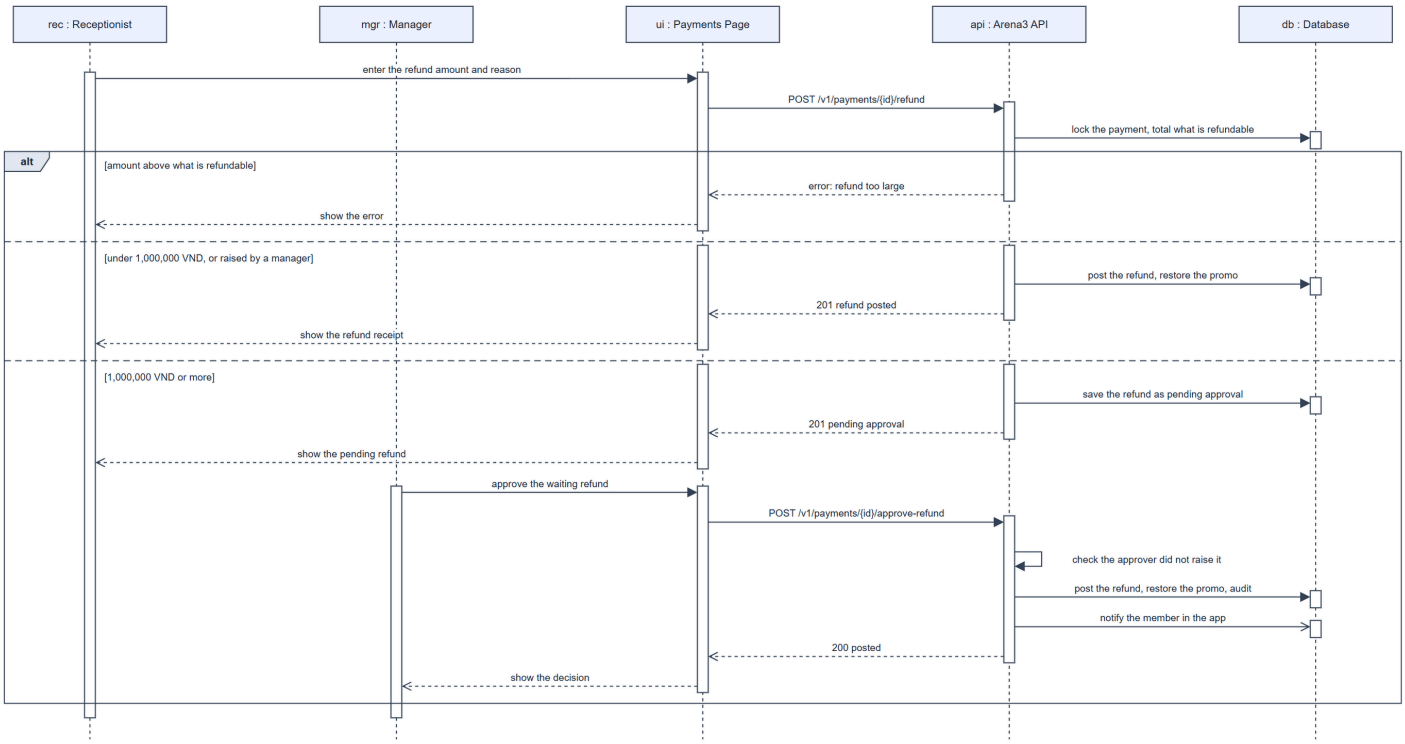


Figure 26. *Sequence diagram: raise and approve a refund*

The activity diagram in Figure 5 (I.3.4) shows the approval branch.

2.25 UC-25 Check in a court booking

Field	Detail
UC ID and name	UC-25 Check in a court booking
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Member
Trigger	The customer arrives for a booking and the receptionist opens Courts (/desk/courts).
Description	The receptionist checks a court booking in on the live court map, inside the check-in window.
Preconditions	The receptionist is signed in. The booking is confirmed.
Postconditions	The booking is checked in; a booking missed past the window becomes a no-show.
Priority	Must have
Frequency of use	Many times a day
Business rules	BR-11

Normal flow

1. The receptionist opens the live court map for today.
2. The system shows each court with its current and next bookings and their payment status.
3. The receptionist selects the booking and chooses Check in.
4. The system checks the window and records the check-in time.

Alternative flows

- 3a. The booking still has a balance: the receptionist collects it first (UC-22).
- 3b. The member shows a booking QR code: the receptionist scans it to open the booking.

Exceptions

- 3c. The booking is not confirmed: CONFLICT_STATE.
- 4a. The time is outside the window of 15 minutes before to 10 minutes after the start: BR-11.

2.26 UC-26 Check a member in at the gate

Field	Detail
UC ID and name	UC-26 Check a member in at the gate
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Member, who shows the gate pass
Trigger	The member shows the gate pass QR at the front desk.
Description	The receptionist admits a member and the system records the visit, flagging any override.
Preconditions	The receptionist is signed in on Gate (/desk/gate). The member opens Gate pass (/app/pass).

Field	Detail
Postconditions	One visit row exists for the member within the 120-minute window. Any override is flagged and audited.
Priority	Must have
Frequency of use	Hundreds of times a day
Business rules	BR-30, BR-31, BR-32, BR-33
Other information	Warnings cover a frozen plan, an expired plan, a plan ending within seven days and a plan with no sessions left. Walking through the gate never spends a session (BR-32) and never fills a class register (BR-33).

Normal flow

1. The member opens Gate pass, which shows a short-lived QR code.
2. The receptionist scans the code on the Gate screen.
3. The system checks the code is a valid member code that has not been used.
4. The system checks today's entitlement: a usable plan, a court booking or a class session today.
5. The system records the check-in time, returns any warnings and shows a green pass.

Alternative flows

- 2a. The code cannot be scanned: the receptionist looks the member up by name, phone or member code and picks why: no phone, dead battery, QR failed, forgot, or other.
- 5a. The member is not entitled: the system lists the reasons and asks for an override.
- 5b. The receptionist picks a reason: expired but allowed, manager approval, guest pass, renewing, or other. The visit is flagged and audited, and the screen shows an amber pass (BR-31).
- 5c. The member already checked in within 120 minutes: the system returns that visit instead of writing a second (BR-31).

Exceptions

- 3a. The code is invalid or already used: BR-30, and the member refreshes the code.
- 3b. The desk's own self check-in code is scanned at the desk: BR-30.

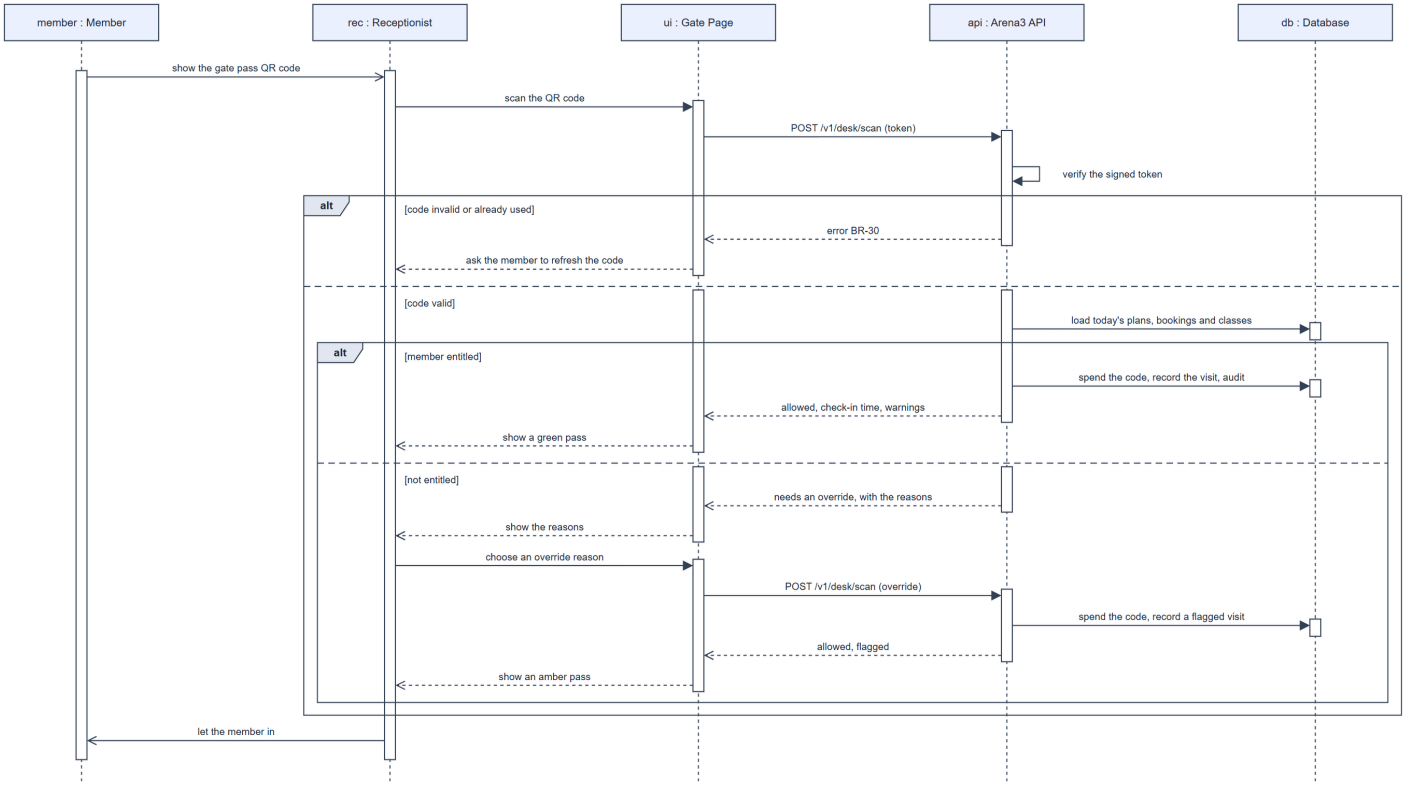


Figure 27. *Sequence diagram: check a member in at the gate*

2.27 UC-27 Rent out gear and record its return

Field	Detail
UC ID and name	UC-27 Rent out gear and record its return
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Guest or member
Trigger	The customer asks for gear at the desk, and the receptionist opens Gear (/desk/gear).
Description	The receptionist rents out gear, charges the hire to the till with an e-invoice, and records the return.
Preconditions	The receptionist is signed in with an open shift. The gear item exists.
Postconditions	Rent: stock is lowered, a loan due in three hours is open, the hire is posted to the shift and an e-invoice is issued. Return: the loan is closed and stock is restored.
Priority	Should have
Frequency of use	Daily
Business rules	BR-44, BR-50, BR-51

Normal flow

1. The receptionist selects the gear item and the quantity (1 to 99).
2. The receptionist picks the member, or enters a guest's phone, and the payment method.
3. The system checks the stock and the open shift.
4. The system lowers the stock and opens the loan, due back in three hours.
5. The system posts the hire (rent × quantity) to the shift, issues the e-invoice and sends the receipt.
6. When the gear comes back, the receptionist selects Return; the system closes the loan and restores the stock.

Alternative flows

- 5a. The item's rent is 0đ: the loan opens without a payment or invoice.

Exceptions

- 2a. The member account is not active, or the guest phone is invalid: VALIDATION.
- 3a. The quantity is more than the stock: BR-50.
- 3b. No open shift: BR-44.
- 6a. The loan is already returned: CONFLICT_STATE.

2.28 UC-28 Handle maintenance work orders

Field	Detail
UC ID and name	UC-28 Handle maintenance work orders
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist, Manager

Field	Detail
Secondary actors	None
Trigger	Staff notice a fault and open Maintenance (/desk/maintenance).
Description	A receptionist reports a fault as a work order, and a manager tracks it to done with its cost.
Preconditions	The user is signed in as a receptionist or a manager.
Postconditions	A work order exists with its priority, status and, when done, its cost. An audit row is written.
Priority	Should have
Frequency of use	Weekly
Business rules	BR-08, BR-53

Normal flow

1. The receptionist enters a title, the court or area, and a priority: low, normal or urgent.
2. The system opens the work order.
3. The manager opens the list, filtered by status.
4. The manager moves the order to in progress, then to done, and records the cost.
5. The system stores each change.

Alternative flows

- 1a. The fault takes a court out of use: the manager sets the court to maintenance, so no booking can use it (BR-08).

Exceptions

- 1b. The title is empty or the priority is unknown: VALIDATION.
- 4a. The cost is negative: VALIDATION.

2.29 UC-29 Reply to a support request

Field	Detail
UC ID and name	UC-29 Reply to a support request
Created by / date	Arena3 team / 2026-10-12
Primary actor	Receptionist
Secondary actors	Member, who receives the reply
Trigger	A support request appears in the desk queue on the desk home (/desk).
Description	The receptionist answers a member's support request, which closes it and notifies the member.
Preconditions	The receptionist is signed in. An open request exists.
Postconditions	The request is closed with the reply, the member is notified, and an audit row is written.
Priority	Should have
Frequency of use	Daily
Business rules	BR-53
Other information	A member who writes again opens a new request, so an answered request never stays in the open queue.

Normal flow

1. The receptionist opens the queue of open requests, oldest first.
2. The receptionist reads a request and writes a reply.
3. The system saves the reply, closes the request and notifies the member.

Alternative flows

- 1a. The receptionist shows all requests to read back an earlier answer.
- 2a. The request needs no reply: the receptionist closes it.

Exceptions

- 3a. The reply is empty: VALIDATION.

2.30 UC-30 View the teaching schedule and rosters

Field	Detail
UC ID and name	UC-30 View the teaching schedule and rosters
Created by / date	Arena3 team / 2026-10-12
Primary actor	Coach
Secondary actors	None
Trigger	The coach opens the coach home (/coach) or My classes (/coach/classes).
Description	The coach sees the week's sessions and the roster of each class taught.
Preconditions	The coach is signed in.
Postconditions	No data changes.
Priority	Must have
Frequency of use	Daily
Business rules	BR-45, BR-46

Normal flow

1. The coach opens the coach home.
2. The system lists the coach's sessions for the week, with court and time.
3. The coach opens a class.
4. The system shows the roster with each student's level and attendance.
5. The coach opens a student to see the profile (UC-34).

Exceptions

- 3a. The coach opens a class the coach neither leads nor assists: FORBIDDEN (BR-45).

2.31 UC-31 Take session attendance

Field	Detail
UC ID and name	UC-31 Take session attendance
Created by / date	Arena3 team / 2026-10-12

Field	Detail
Primary actor	Coach
Secondary actors	Manager, who corrects closed registers
Trigger	The session starts and the coach opens Attendance (/coach/attendance).
Description	The coach marks each student present, late, absent or excused for a session.
Preconditions	The coach leads or assists the class. The register is not locked.
Postconditions	The register is saved. Three absences in a row raise an alert.
Priority	Must have
Frequency of use	Every session
Business rules	BR-33, BR-34, BR-35, BR-36, BR-45

Normal flow

1. The coach selects the session.
2. The system lists the enrolled students.
3. The coach marks each one present, late, absent or excused.
4. The system saves the register and checks for absence streaks.

Alternative flows

- 1a. A manager corrects a locked register: the manager enters a reason, and the change is audited (BR-35).
- 4a. A student reaches three absences in a row: the system alerts the coach and the managers (BR-36).

Exceptions

- 1b. More than 2 hours have passed since the session: BR-35.
- 1c. The session belongs to another coach's class: FORBIDDEN (BR-45).
- 3a. A result other than present, late, absent or excused: VALIDATION.

2.32 UC-32 Plan a training session

Field	Detail
UC ID and name	UC-32 Plan a training session
Created by / date	Arena3 team / 2026-10-12
Primary actor	Coach
Secondary actors	Gemini AI service
Trigger	The coach opens a session in Training plans (/coach/classes).
Description	The coach builds the plan of a session from blocks, a saved template or an AI suggestion.
Preconditions	The coach teaches the class. The training module is switched on.
Postconditions	The session plan is saved as a new version.
Priority	Should have
Frequency of use	Weekly

Field	Detail
Business rules	BR-45, BR-47, BR-48, BR-49, BR-52

Normal flow

1. The coach opens the session and selects Plan.
2. The coach adds blocks, each with a title, type, intensity, minutes and description.
3. The system checks the blocks and saves the plan.

Alternative flows

- 2a. The coach starts from a saved template, then edits the blocks.
- 2b. The coach asks for an AI suggestion: the system sends the class sport and level to Gemini and fills the blocks, which the coach reviews.
- 3a. The coach saves the plan as a template, or copies a week of plans to the next week.

Exceptions

- 1a. The training module is switched off: BR-52.
- 3b. A block breaks BR-49: VALIDATION, and the block is marked.
- 3c. The session already has results: BR-48, and the coach copies the plan instead.

2.33 UC-33 Record session results and homework

Field	Detail
UC ID and name	UC-33 Record session results and homework
Created by / date	Arena3 team / 2026-10-12
Primary actor	Coach
Secondary actors	Member, who receives homework
Trigger	The session ends and the coach opens it in Training plans.
Description	The coach records how each student did in the session and sets homework.
Preconditions	The coach teaches the class. The training module is switched on.
Postconditions	Results and homework are saved; students see them in Training.
Priority	Should have
Frequency of use	Every session
Business rules	BR-45, BR-47, BR-48

Normal flow

1. The coach opens the session results.
2. The coach records a score and a note for each student.
3. The system saves the results.
4. The coach sets homework with a title, a checklist and a due date, for the class or one student.
5. The system saves the homework and notifies the students.

Exceptions

- 3a. The session belongs to another coach's class: FORBIDDEN (BR-45).
- 4a. The title is longer than 120 characters, the checklist has more than 20 items, or the due date is past: VALIDATION.

2.34 UC-34 Assess a student's progress

Field	Detail
UC ID and name	UC-34 Assess a student's progress
Created by / date	Arena3 team / 2026-10-12
Primary actor	Coach
Secondary actors	Member
Trigger	The coach opens a student profile (/coach/student/$id).
Description	The coach sets a student's level, writes a 2- or 4-week review and keeps private notes.
Preconditions	The student is in a class the coach teaches.
Postconditions	The level, review or note is saved and the student sees the review.
Priority	Should have
Frequency of use	Weekly
Business rules	BR-45, BR-46

Normal flow

1. The coach opens the student profile.
2. The system shows level, attendance, results, homework progress and health notes in scope.
3. The coach writes a review covering 2 or 4 weeks with scores from 1 to 5.
4. The system saves the review and shows it to the student.

Alternative flows

- 3a. The coach changes the student's level.
- 3b. The coach adds a private note of under 2,000 characters.

Exceptions

- 1a. The student is not in the coach's classes: FORBIDDEN (BR-45).
- 3c. A score outside 1 to 5, or a period other than 2 or 4 weeks: VALIDATION.

2.35 UC-35 View monthly earnings

Field	Detail
UC ID and name	UC-35 View monthly earnings
Created by / date	Arena3 team / 2026-10-12
Primary actor	Coach
Secondary actors	None
Trigger	The coach opens Earnings (/coach/earnings).

Field	Detail
Description	The coach sees the sessions taught and the commission earned for a month, and whether it is paid.
Preconditions	The coach is signed in.
Postconditions	No data changes.
Priority	Could have
Frequency of use	Monthly
Business rules	None

Normal flow

1. The coach opens Earnings and picks a month.
2. The system shows the sessions taught, the students taught, the rate and the amount.
3. The system shows whether the month is open, closed or paid.

Exceptions

- 1a. The month has no sessions: the system shows zero.

2.36 UC-36 Manage user accounts

Field	Detail
UC ID and name	UC-36 Manage user accounts
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	None
Trigger	The manager opens Staff (/manager/staff) or Members (/manager/members).
Description	The manager creates and maintains staff accounts and looks after member accounts.
Preconditions	The manager is signed in.
Postconditions	An account is created or changed, sessions may be revoked, and an audit row is written.
Priority	Must have
Frequency of use	Weekly
Business rules	BR-01, BR-25, BR-53

Normal flow

1. The manager selects Add staff.
2. The manager enters name, phone, email and role: receptionist or coach, with the sports for a coach.
3. The system creates the account with a temporary password.
4. The manager passes the temporary password to the new staff member, who changes it on first sign-in.

Alternative flows

- 1a. The manager locks or unlocks an account, resets its password, or signs it out everywhere.
- 1b. The manager searches the member directory by status, sport or plan and opens a member.

Exceptions

- 1c. The manager tries to change their own account here: FORBIDDEN.
- 1d. A coach with running classes is given another role: CONFLICT_STATE.
- 2a. The phone or email already has an account: BR-01.
- 2b. A coach has no sport: VALIDATION.

2.37 UC-37 Manage membership plans and prices

Field	Detail
UC ID and name	UC-37 Manage membership plans and prices
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	None
Trigger	The manager opens Plans (/manager/plans) or Prices (/manager/prices).
Description	The manager sets up the plans on sale and the court price rules.
Preconditions	The manager is signed in.
Postconditions	Plans or price rules are saved and apply to new orders and bookings. An audit row is written.
Priority	Must have
Frequency of use	Monthly
Business rules	BR-16, BR-17, BR-37, BR-38, BR-53

Normal flow

1. The manager creates or edits a plan: name, sport, length, sessions, court hours, discount and price.
2. The system checks and saves the plan.
3. The manager opens Prices and edits the price rules by sport, court, weekday and hour.
4. The system saves the rules.

Alternative flows

- 2a. The manager withdraws a plan: it stays on existing members but can no longer be ordered (BR-16).

Exceptions

- 2b. The discount is not a whole number from 0 to 100: BR-17.
- 4a. A rule has no price, or overlaps another rule: VALIDATION.

2.38 UC-38 Manage promo codes

Field	Detail
UC ID and name	UC-38 Manage promo codes
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	None
Trigger	The manager opens Promo codes (/manager/promos).
Description	The manager creates promo codes with their scope, limits and dates, and pauses them.
Preconditions	The manager is signed in.
Postconditions	The code is saved or paused. An audit row is written.
Priority	Should have
Frequency of use	Monthly
Business rules	BR-40, BR-41, BR-42, BR-43

Normal flow

1. The manager enters the code, a percent or fixed amount, the scope, the dates and the limits.
2. The system checks and saves the code.
3. The manager views how many times each code has been used.

Alternative flows

- 3a. The manager pauses or resumes a code.

Exceptions

- 2a. The code is not 3 to 24 characters, a percent is over 100, or a fixed amount is not a multiple of 1,000đ: VALIDATION.
- 2b. The manager renames a code that has been used: BR-43.

2.39 UC-39 Schedule classes and sessions

Field	Detail
UC ID and name	UC-39 Schedule classes and sessions
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	Coach, who is assigned
Trigger	The manager opens Classes (/manager/classes).
Description	The manager creates a class with its coach, court, schedule and capacity, opens it, and changes or cancels sessions.
Preconditions	The manager is signed in.
Postconditions	The class and its sessions exist and hold their court times. Enrolled members are notified of changes.
Priority	Must have

Field	Detail
Frequency of use	Weekly
Business rules	BR-08, BR-21, BR-25, BR-26, BR-27, BR-28, BR-29

Normal flow

1. The manager enters the sport, coach, court, weekdays, time, length (60, 120 or 180 minutes), capacity and dates.
2. The system checks the coach and the court times and creates the sessions.
3. The manager publishes the class, which opens it for enrolment.

Alternative flows

- 3a. The manager moves a session to a future time, with a reason when it is under 12 hours away (BR-27).
- 3b. The manager cancels a session with a reason; session-based plans get the session back once (BR-29).
- 3c. The manager hands the class to a substitute coach from a date onward (BR-28).
- 3d. The manager changes the capacity.

Exceptions

- 2a. The coach is inactive or not qualified for the sport: BR-25.
- 2b. The court or the coach is busy at a session time: BR-26 or CONFLICT_SLOT.
- 3e. The capacity is outside 1 to 200, or below the number enrolled: BR-21.

2.40 UC-40 Configure the centre

Field	Detail
UC ID and name	UC-40 Configure the centre
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	None
Trigger	The manager opens Centre settings (/manager/settings).
Description	The manager sets the centre's hours, booking limits, refund tiers, check-in and other settings, and switches modules on or off.
Preconditions	The manager is signed in.
Postconditions	The settings apply to every later request. An audit row is written.
Priority	Must have
Frequency of use	Rarely
Business rules	BR-52, BR-53

Normal flow

1. The manager opens Centre settings.

2. The system shows the settings in groups with their current values.
3. The manager edits values and saves.
4. The system checks each value's range and stores them.

Alternative flows

- 3a. The manager switches a module off: every route of that module answers 403 with BR-52 until it is switched on.

Exceptions

- 4a. A value is out of range: VALIDATION, naming the setting.

2.41 UC-41 View reports and export data

Field	Detail
UC ID and name	UC-41 View reports and export data
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	None
Trigger	The manager opens Reports (/manager), Attendance (/manager/attendance) or Invoices (/manager/invoices).
Description	The manager reviews revenue, occupancy, capacity, members and attendance, and exports reports and e-invoices.
Preconditions	The manager is signed in.
Postconditions	No data changes; an export file is downloaded.
Priority	Must have
Frequency of use	Daily
Business rules	BR-35, BR-38

Normal flow

1. The manager picks a report and a date range.
2. The system builds the report from the same queries as the screens.
3. The manager selects Export and a format: xlsx or pdf.
4. The system builds the file on request and the browser downloads it.

Alternative flows

- 1a. The manager opens Invoices and exports the e-invoices of a period as XML.

Exceptions

- 2a. The date range is invalid: VALIDATION.

2.42 UC-42 Approve or reject a refund

Field	Detail
UC ID and name	UC-42 Approve or reject a refund
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	Receptionist, who raised the refund; Member, who is notified
Trigger	A refund pending approval appears in Refund sign-off (/desk/payments).
Description	A manager other than the person who raised a large refund approves or rejects it.
Preconditions	The manager is signed in. The refund is pending approval.
Postconditions	Approve: the refund is posted and the member notified. Reject: it is closed with a reason. An audit row is written.
Priority	Must have
Frequency of use	A few times a week
Business rules	BR-53

Normal flow

1. The manager opens the pending refunds.
2. The manager reviews the payment, the amount and the reason.
3. The manager selects Approve.
4. The system posts the refund and notifies the member.

Alternative flows

- 3a. The manager selects Reject and gives a reason; the system closes the refund and notifies the member.

Exceptions

- 3b. The manager raised this refund: FORBIDDEN, and another manager must sign it off.
- 3c. Another manager decided it first: CONFLICT_STATE.

2.43 UC-43 Close and pay coach commission

Field	Detail
UC ID and name	UC-43 Close and pay coach commission
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	Coach, who is paid
Trigger	A month ends and the manager opens Commission (/manager/commission).
Description	The manager sets coach rates, closes a finished month so its figures are frozen, and marks each coach paid.
Preconditions	The manager is signed in.
Postconditions	Close: one payout row per coach with sessions. Pay: the payout is paid. An audit row is written.

Field	Detail
Priority	Should have
Frequency of use	Monthly
Business rules	BR-53

Normal flow

1. The manager picks a month and reviews each coach's sessions, students and amount.
2. The manager selects Close month; the system copies the figures into payout rows.
3. The manager pays a coach outside the system and selects Mark paid.

Alternative flows

- 1a. The manager changes a coach's rate; closed months keep their figures.

Exceptions

- 2a. The month has not ended: CONFLICT_STATE.
- 3a. The month is not closed, or already paid: CONFLICT_STATE.

2.44 UC-44 Review the audit log

Field	Detail
UC ID and name	UC-44 Review the audit log
Created by / date	Arena3 team / 2026-10-12
Primary actor	Manager
Secondary actors	None
Trigger	The manager opens Audit log (/manager/audit).
Description	The manager reviews who changed what and when, to investigate a dispute or a mistake.
Preconditions	The manager is signed in.
Postconditions	No data changes.
Priority	Must have
Frequency of use	Weekly
Business rules	BR-53

Normal flow

1. The manager opens Audit log.
2. The system lists audit rows, newest first, with actor, action, record and time.
3. The manager filters by actor, action or date.
4. The manager opens a row to see the values before and after.

Exceptions

- 3a. No rows match: the system shows an empty list.

IV. Non-functional requirements

Arena 3 must keep money and schedules correct under concurrency, respond quickly on a phone, and protect member data. This section names the external interfaces and the quality attributes that make this so.

1. External interfaces

1.1 User interfaces

Arena 3 is one responsive web application with four portals, and every screen works from a 375 px phone width.

- The interface is available in English and Vietnamese, and the user switches language in the header.
- Running text uses Be Vietnam Pro, which renders Vietnamese diacritics correctly. Headings use Newsreader and figures use Anton.
- Amounts are written in VND with a lowercase đ, for example 1,500,000đ.
- A refusal is shown as a sentence that says what to do instead. The error code travels with it for support.
- Motion is decorative only, and no action is discoverable only through animation.

1.2 Software interfaces

The application talks to five outside systems, and each has a defined fallback.

Interface	Purpose	Protocol and data	Failure behaviour
Arena 3 API (`/v1`)	Every screen reads and writes through it	JSON over HTTPS with a bearer token; served inside the web app	Errors return `{ code, message }`, plus `field` or `br` where one applies
payOS	Payment links, VietQR codes, link status and payment webhooks	HTTPS REST; the webhook is signed with the checksum key	A payment posts only when payOS confirms it, and the amount must equal the amount due
Google Gemini (`gemini-2.5-flash`)	Answers in the AI assistant	HTTPS; the prompt carries centre data only, never member data	12-second timeout, then xAI Grok, then rule-based answers
Email (SMTP through nodemailer)	One-time codes and notifications	SMTP with the centre's sender address	Codes are sent at once; notifications are queued and retried up to five times
PostgreSQL 16	All records	Neon in production; embedded PGLite when no `DATABASE_URL` is set	Schema changes are forward-only numbered migrations, each run in one transaction

The application is hosted on Vercel. Background jobs start with the first API request and run in a loop every 15 seconds.

1.3 API error codes

Every API error uses one of these codes, so a screen can react to the code rather than parse the text.

HTTP	Code	Meaning
400	`VALIDATION`	A value is malformed; `field` names the input to correct
401	`UNAUTHENTICATED`	No session, or the session has expired
403	`FORBIDDEN`	Wrong role, or another member's data. With `br: "BR-52"`, the module is switched off
404	`NOT_FOUND`	The record does not exist
409	`CONFLICT_SLOT`	Another person took that court time first
409	`HOLD_EXPIRED`	The five-minute hold ran out
409	`CONFLICT_STATE`	The record is in the wrong state for this action
422	`BR_VIOLATION`	A named business rule refused the request; `br` carries the rule code
422	`VALIDATION`	The request is well formed but not acceptable in the current state
429	`RATE_LIMITED`	Too many attempts in a short time

2. Quality attributes

2.1 Correctness and reliability

Two bookings can never own the same court time, and money never disagrees with the schedule.

ID	Requirement
NFR-01	No two bookings may own the same court time, under any concurrency. Both booking paths lock the court row, and a database trigger rejects any overlap.
NFR-02	No coach may be scheduled into two overlapping sessions.
NFR-03	A confirmed booking always has its payment and invoice, written in the same transaction.
NFR-08	Any request that takes money or changes a schedule accepts an `Idempotency-Key`. A replay returns the first result and never charges twice.
NFR-09	With no `DATABASE_URL`, the application migrates and seeds an embedded PGLite database, so a fresh copy runs offline.
NFR-10	Losing the assistant's model degrades only the assistant, which falls back to rule-based answers.

2.2 Performance

The screens staff and members open most often must cost as few database round trips as possible.

ID	Requirement
NFR-04	`GET /v1/me` issues its independent reads concurrently, so a sign-in costs one round trip rather than seven.
NFR-05	Read-only endpoints open no transaction, which saves two round trips to a remote database.
NFR-06	The front desk badge asks for a count only (`?brief=1`), which runs one aggregate query.
NFR-07	The job loop first checks, in one query, which jobs are due. A pass with nothing due costs one query.

2.3 Usability

ID	Requirement
NFR-11	Every screen works at a width of 375 px.
NFR-12	A refusal tells the user what to do instead, as a sentence rather than a code.
NFR-13	Text renders Vietnamese diacritics correctly, and currency is never upper-cased (1,500,000đ, never 1,500,000Đ).
NFR-14	Motion is decorative and never carries meaning on its own.

2.4 Auditability

ID	Requirement
NFR-15	Every staff-only change writes an audit row naming the actor, action, record and time. This covers refunds, reconciliation, shift closes, settings, sessions, classes, staff accounts, attendance corrections and payment links. No endpoint deletes an audit row.
NFR-16	Member, payment and invoice codes come from a counter table and have no gaps within each kind.

2.5 Security and privacy

Authorisation is decided on the server for every request, and member data stays inside the centre.

ID	Requirement
NFR-17	Passwords are stored only as salted hashes and are never written to logs.
NFR-18	Session tokens are stored as SHA-256 hashes. Member sessions last seven days and staff sessions twelve hours.
NFR-19	Every request is authorised on the server. Hiding a button is not a control.
NFR-20	A member can read only their own bookings, receipts and profile. A read of another member's data returns `403`.
NFR-21	Health notes and guardian details are shown only to those who need them and are never sent to the assistant's model.
NFR-22	In production, one-tap demo sign-in and on-screen codes are off unless `VITE_DEMO_LOGINS` or `OTP_ECHO` is set explicitly.
NFR-23	A switched-off module (F4 training) answers `403` with `BR-52` on every route.
NFR-24	Report exports are built on request from the same queries as the screens, and are not stored.
NFR-25	An online payment is trusted only when read from payOS or carried by a verified webhook signature.
NFR-26	Sign-in locks for 15 minutes after five wrong passwords, and a one-time code after five wrong attempts.

2.6 Maintainability

The requirements in this document are checked by executable tests, so a change that breaks one fails before release.

- `npm test` runs the unit tests for pricing, refunds, check-in windows and other rules.
- `npm run check:api` drives the real API against a real database and checks the money and schedule rules end to end.
- `npm run typecheck` checks the TypeScript types across the application.

V. Requirement appendix

This appendix lists every business rule, configurable setting, shared requirement and user message that the sections above refer to. Each entry matches the running application.

1. Business rules

Rules are grouped by business area and typed by the standard business-rule taxonomy. A Fact states what is true; a Constraint limits what may be done; an Action enabler lets one action start, or makes it require another; a Computation derives a value; an Inference draws a conclusion from other facts. A refused request returns `422 BR_VIOLATION` with the rule code in `br`, so any refusal on screen traces to one row below.

ID	Category	Rule statement	Type	Applies to
BR-01	Accounts and access	Each user account must have a phone number and an email address that no other account uses.	Constraint	UC-02, UC-19, UC-36
BR-02	Accounts and access	A password must contain at least 8 characters, including at least one letter and one digit.	Constraint	UC-02, UC-04, UC-05
BR-03	Accounts and access	A member under 16 years of age must have a guardian's name and phone number on record.	Constraint	UC-02, UC-19
BR-04	Accounts and access	A member account may be created only after the person accepts the terms of use and the data-privacy notice.	Action enabler	UC-02, UC-19
BR-05	Court booking	A member may book a court at most 7 days ahead and may hold at most 2 court slots on the same day.	Constraint	UC-08, UC-21
BR-06	Court booking	A court booking must lie within opening hours, on a court whose status is ready.	Constraint	UC-08, UC-09, UC-21
BR-07	Court booking	A court booking must not start in the past. A walk-in booking must have at least 20 minutes left in its slot.	Constraint	UC-08, UC-21
BR-08	Court booking	Each court time must have exactly one owner: a booking, a class session or a maintenance block.	Constraint	UC-08, UC-21, UC-39
BR-09	Court booking	A slot that overlaps one of the member's classes may be held only after the member confirms the clash.	Action enabler	UC-08, UC-09
BR-10	Court booking	A replacement hold must pass every booking check before the hold it replaces is released.	Constraint	UC-08
BR-11	Court booking	Court check-in is allowed from 15 minutes before the start until 10 minutes after it. A booking not checked in by then is a no-show.	Inference	UC-25
BR-12	Membership plans	A plan grants its benefits only between its start date and its end date.	Inference	UC-08, UC-13, UC-20
BR-13	Membership plans	A court booking paid from plan hours requires enough court hours left on that plan.	Constraint	UC-08
BR-14	Membership plans	A frozen plan must be resumed before it is frozen again or paid for.	Constraint	UC-10, UC-20

ID	Category	Rule statement	Type	Applies to
BR-15	Membership plans	A plan may be frozen for no more than the yearly freeze allowance (30 days by default).	Constraint	UC-20
BR-16	Membership plans	A withdrawn plan must not be ordered or sold.	Constraint	UC-10, UC-20, UC-37
BR-17	Membership plans	A plan discount is a whole-number percentage from 0 to 100.	Constraint	UC-37
BR-18	Classes and enrolment	A member may enrol only in a class whose status is open.	Constraint	UC-13
BR-19	Classes and enrolment	Class enrolment requires an active plan that covers the class sport.	Action enabler	UC-13, UC-14
BR-20	Classes and enrolment	Enrolment through a session-based plan requires sessions left on that plan.	Constraint	UC-13
BR-21	Classes and enrolment	Enrolments must not exceed class capacity, and capacity must not be set below the number already enrolled.	Constraint	UC-13, UC-14, UC-39
BR-22	Classes and enrolment	A member must not hold a duplicate enrolment or two enrolments whose sessions overlap.	Constraint	UC-13
BR-23	Classes and enrolment	A waitlist offer expires 2 hours after it is sent; the seat then passes to the next member in line.	Inference	UC-14
BR-24	Classes and enrolment	A member may leave a class only up to 4 hours before its next session.	Constraint	UC-15
BR-25	Classes and enrolment	A class coach must hold an active account and be qualified for the class sport.	Constraint	UC-39
BR-26	Classes and enrolment	A session must not be moved, or its coach changed, to a time when the court or the coach is already busy.	Constraint	UC-39
BR-27	Classes and enrolment	Moving a session less than 12 hours before its original start requires a reason.	Constraint	UC-39
BR-28	Classes and enrolment	A substitute coach takes over a class from the date of the change onward; past sessions keep their coach.	Inference	UC-39
BR-29	Classes and enrolment	When the centre cancels a session, a member on a session-based plan receives that session back once.	Computation	UC-39
BR-30	Check-in and attendance	A gate code must be a valid member code that has not been used before.	Constraint	UC-12, UC-26
BR-31	Check-in and attendance	A member with no entitlement today is admitted only with an override reason, and the visit is flagged. A repeat scan within 120 minutes returns the first visit.	Action enabler	UC-26
BR-32	Check-in and attendance	A gate check-in never spends a session or changes any balance.	Constraint	UC-12, UC-26
BR-33	Check-in and attendance	A gate check-in never marks class attendance; only the session register does.	Constraint	UC-26, UC-31
BR-34	Check-in and attendance	A student is marked late only by the coach in the session register.	Constraint	UC-31
BR-35	Check-in and attendance	A session register locks 2 hours after the session; afterwards only a manager may correct it, and must give a reason.	Constraint	UC-31, UC-41

ID	Category	Rule statement	Type	Applies to
BR-36	Check-in and attendance	Three absences in a row alert the coach and the managers; an excused absence breaks the streak.	Inference	UC-19, UC-31
BR-37	Pricing, promotions and payment	A price is quoted only from a matching price rule; the system never invents a price.	Constraint	UC-08, UC-21, UC-37
BR-38	Pricing, promotions and payment	Every price shown includes VAT.	Fact	UC-37, UC-41
BR-39	Pricing, promotions and payment	Money is rounded once, half up, to the nearest 1,000đ after every discount is applied.	Computation	UC-08, UC-20, UC-22
BR-40	Pricing, promotions and payment	A promo code must be active, within its dates, and match the order's scope, sport, plan and minimum spend.	Constraint	UC-08, UC-10, UC-38
BR-41	Pricing, promotions and payment	A promo code must not be used beyond its total limit or its per-member limit.	Constraint	UC-08, UC-38
BR-42	Pricing, promotions and payment	A promo code that does not stack never combines with a plan discount; the larger of the two applies.	Computation	UC-08
BR-43	Pricing, promotions and payment	A promo code that has been used must keep its code; staff pause it and create a new one instead.	Constraint	UC-38
BR-44	Pricing, promotions and payment	Money may be taken at the desk only within the receptionist's open cashier shift.	Action enabler	UC-18, UC-20, UC-21, UC-22, UC-23, UC-24, UC-27
BR-45	Coaching and training	A coach may act only on classes that the coach leads or assists.	Constraint	UC-30, UC-31, UC-32, UC-33, UC-34
BR-46	Coaching and training	A student's health notes are shown only to staff whose role or class gives them a need to know.	Constraint	UC-19, UC-34
BR-47	Coaching and training	Training records never take money or change a plan balance.	Constraint	UC-32, UC-33
BR-48	Coaching and training	A training session that has results cannot change or add a plan; the coach copies the plan instead.	Constraint	UC-32, UC-33
BR-49	Coaching and training	A training plan has 1 to 12 blocks, and each block carries a type, an intensity, a description and a length of 1 to 180 minutes.	Constraint	UC-32
BR-50	Equipment and system	Gear may not be rented out beyond the stock on hand.	Constraint	UC-27
BR-51	Equipment and system	Gear hire is charged to the open cashier shift and issues an e-invoice; a return puts the gear back in stock.	Action enabler	UC-27
BR-52	Equipment and system	A module that a manager has switched off answers every request with 403 and BR-52.	Constraint	UC-40
BR-53	Equipment and system	Every change to money, plans, prices, accounts or settings writes an audit row, and no audit row is ever deleted.	Action enabler	UC-44

Version 2.1 renumbered the rules by category, from 75 codes to 53. Codes that only described a screen, or repeated another rule, were folded into the rule they belong to, and the API returns the new

codes. BR-49, BR-51 and BR-53 state rules the application already enforced without a code.

2. Centre settings

Managers change these values on Centre settings (`/manager/settings`), and the rules above read them on every request.

Group	Setting	Default
Locale	Time zone; currency	Asia/Ho_Chi_Minh; VND (both fixed)
Opening hours	Open; close; slot length	06:00; 22:00; 60 minutes
Booking	Hold time; days ahead; slots per member per day	5 minutes; 7 days; 2
Cancellation	Court booking notice; class notice	2 hours; 4 hours
Court refunds by notice	24 hours or more; 4 to 24 hours; under 4 hours	100%; 50%; 0%
Check-in	Court check-in window; no-show grace	15 minutes before; 10 minutes after
Deposit	Deposit share; peak slots only	30%; yes
Fixed weekly bookings	Length; discount	4 to 12 weeks; 10%
Money	Refund limit without a manager; VAT rate; rounding	1,000,000đ; 0%; nearest 1,000đ
Members	Freeze days a year; minor age; waitlist offer	30 days; under 16; 2 hours
Transfers	Bank transfer hold	120 minutes
Loyalty and day pass	Spend per point; value per point; day pass price	10,000đ; 100đ; 60,000đ
Gate	Self check-in; repeat-scan window	Off; 120 minutes

3. Common requirements

These requirements apply to every function, so the use cases do not repeat them.

Area	Requirement
Idempotency	Requests that take money or change a schedule accept an `Idempotency-Key`; a replay returns the first result.
Audit	Every staff-only change writes an audit row, and no endpoint deletes one.
Money	Amounts are whole VND, rounded once to the nearest 1,000đ after discounts. Payments are never deleted; a refund is a new negative payment.
Codes	Member, payment and invoice codes have no gaps within each kind.
Time	All times are stored with a time zone and shown in Asia/Ho_Chi_Minh time.
Sessions	Member sessions last seven days and staff sessions twelve hours.
Lockout	Five wrong passwords, or five wrong one-time codes, lock that path for 15 minutes.
Rate limits	Assistant questions: 800 characters each, 20 a minute. Plan orders: 3 a minute, 10 an hour.
Notifications	Notifications are queued and sent in batches of 50, with up to five attempts each. Plan reminders go 7, 3 and 0 days before the end date.
Background jobs	The jobs listed in III.1.4 run in one loop every 15 seconds.
Language	Every screen is available in English and Vietnamese, and Vietnamese diacritics render correctly.
Layout	Every screen works at a width of 375 px.

4. Application messages

Each message below is the text the application shows today. The code travels alongside it so support can trace the refusal.

#	Code	Type	Message
MSG01	BR-01	Error	That phone or email already has an account.
MSG02	BR-02	Error	Password needs at least 8 characters, with letters and numbers.
MSG03	RATE_LIMITED	Error	Too many OTP attempts — locked for 15 minutes.
MSG04	RATE_LIMITED	Error	5 wrong passwords — locked for 15 minutes.
MSG05	BR-05	Error	You can book at most 7 days ahead.
MSG06	BR-09	Confirmation	This slot clashes with a class you are in. Confirm to hold it anyway.
MSG07	CONFLICT_SLOT	Error	Someone just took that slot.
MSG08	HOLD_EXPIRED	Error	Your hold has expired.
MSG09	BR-37	Error	There is no price for that slot yet — ask the desk.
MSG10	BR-40	Error	That code has expired.
MSG11	BR-19	Error	You need an active plan covering this sport.
MSG12	BR-23	Error	That offer has expired.
MSG13	BR-44	Error	Open a till shift before taking payment.
MSG14	BR-30	Error	That code was already used — ask the member to refresh it.
MSG15	VALIDATION	Error	This payment has already been refunded in full.
MSG16	FORBIDDEN	Error	You raised this refund, so someone else has to sign it off.
MSG17	CONFLICT_STATE	Error	That refund is no longer waiting for a decision.
MSG18	BR-35	Error	This register closed 2 hours after the session. Ask a manager to correct it.
MSG19	FORBIDDEN	Error	You do not have permission.
MSG20	RATE_LIMITED	Error	Too many attempts — try again later.
MSG21	BR-50	Error	Only {stock} {item} left — you asked for {quantity}.